The Conjectures of Ashoke Sen

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Ashoke Sen made many conjectures about string theory over thirty years: that a certain symmetry of the theory is exact, that two magnetic monopoles form a bound state, that an unstable brane decays to a vacuum with no open strings and leaves behind a dust with no pressure, that a path integral counts the states of a black hole, that the number of states of a dyon is positive. We went through all of them, thirty-six statements in all, read the papers in which they were made and the papers that tested them, and checked every reference. We say for each one whether it has been proved, whether it has passed every test without a proof, whether it was stretched beyond the case where it holds, or whether it is false. Most of the conjectures are true. Two are false: the positivity of the dyon count fails in two orbifold models, where we recompute the negative numbers exactly, and the pressureless dust is not the end state of a brane decaying unevenly, where we prove that the dust piles up to infinite density in a finite time. For everything that is still open we say what is missing.

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I. INTRODUCTION

Ashoke Sen has published conjectures for more than thirty years, and most of them were made in the same way: a computation in one regime, a symmetry that carries it to a regime where nothing can be computed, and a statement about that second regime sharp enough to be refuted. The statement that the heterotic string on a six-torus is invariant under $SL(2,\mathbb{Z})$ required a bound state of two monopoles that nobody had seen [112, 113]. The statement that the tachyon of an unstable D-brane rolls to a vacuum without open strings, at an energy equal to the tension of the brane [116, 122], required an exact solution of open string field theory that did not exist for seven years. The statement that the Fourier coefficients of a Siegel modular form have a definite sign on a cone of charges [97] required an argument about the angular momentum of a black hole. These are predictions in the strict sense, and they can be checked.

This paper checks them. We have taken every conjecture, and every claim that was presented as more than a computation, from the papers of Sen and his collaborators that we could verify, and we have asked three questions of each: what exactly was claimed, and under which hypotheses; what has been proved since, by whom, and in which setting; and where a claim is incomplete or false, which step fails. The answers are given in the form of labelled statements. Where a conjecture has become a theorem we state the theorem and name its proof. Where it is confirmed by every test but not proved we say so, and we say what a proof would have to supply. Where it fails we give the counterexample and the hypothesis that breaks, with a computation that anyone can repeat.

The verdicts are of four kinds, and we use them as technical

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terms throughout.[1]

- PROVED: the statement, as made, follows from a published argument that we have read and that is complete within its framework. Where the framework is itself a conjecture (open string field theory as a complete theory of the open string, for instance) we say so.
- CONFIRMED: no proof exists and no test contradicts the statement. We name the tests, or say that there is none, and we name the missing step.
- INCOMPLETE: the statement is correct in the regime where it was derived and has been extended, in the literature or in the statement itself, to a regime where it is not established. We give the corrected statement.
- FALSE AS STATED: there is a counterexample. We give it and we isolate the hypothesis that fails.

Two colours mark the errors in the text. A sentence of Sen's that is fatally wrong, in the sense that the result it states falls and not merely a step of its proof, is printed in burnt orange where we quote or restate it; the correction follows in ordinary type. A sentence that is wrong but not fatal, because it holds in the regime where it was derived and fails only in the regime to which it was extended, or because it asserts as established a step that is open, is marked with a yellow highlight. Nothing else is coloured. Besides the two fatal errors, the yellow marks fall on the Calabi–Yau form of the Sen limit (Section IX), on the Gauss–Bonnet entropy of small black holes above five dimensions (Section X B) and on the claim that localisation derives the Rademacher expansion of the index (Section XI A).

The result of the examination is not what a reader who starts from the suspicion that these conjectures are "irrelevant, unproved, incomplete" would expect, and we record it without softening it in either direction. Of the thirty-six claims of the catalogue, twenty-two are proved, ten are confirmed, two are incomplete, and two are false as stated. For every claim that is not proved we searched the literature that cites it, through October 2026, for a proof, and the verdict records what we found and the step that is missing. Statements that Sen corrected himself in a later paper are not errors of the record and are not listed. The errors are real and we treat them at length. The positivity conjecture for the index of single-centred quarter-BPS dyons [97] fails in the freely acting $\mathbb{Z}_2$ and $\mathbb{Z}_3$ orbifolds of type II string theory on T^6, as Chattopadhyaya and David found [13]; we recompute the indices from the product formula in exact arithmetic (Theorem V.4), we list twenty-one charges in the single-centred cone with negative index, and we trace the failure to the explicit hypothesis of Sen's argument that a duality frame exists in which the black hole carries no hair beyond the fermion zero modes of broken supersymmetry; in these two models no such frame exists [14]. The hypothesis holds in the heterotic string, where the conjecture was made and where it is now a theorem [10, 74], and in the CHL models, where it is confirmed. The second failure is the identification of tachyon matter, the pressureless dust left at the end of the decay of an unstable brane [83–85], as the final state of the decay. The identification is correct for homogeneous decay. For inhomogeneous decay we prove (Theorem VII.4) that a solution of the dust equations of [83] develops a caustic in finite time exactly when its initial profile is not a convex function, so that every bounded profile that is not constant does, with an explicit formula for the time, and that the energy density diverges there; the effective description stops at that time, in agreement with the numerical and perturbative studies [4, 40, 41], and the closed strings emitted by the decaying brane [63] carry away all its energy in low dimension. What remains of the conjecture is a theorem about homogeneous decay and an open question about the inhomogeneous one, and we state both.

The two incomplete claims are the matching of the entropy of small black holes by the Gauss–Bonnet term, which holds in four and five dimensions and fails in six and more, by Sen's own computation [91], and the weak coupling limit of F-theory on a Calabi–Yau threefold [115], whose seven-brane is singular along the orientifold plane [15] and whose existence fails on most bases with non-Higgsable clusters [66, 67]. None of these is an error in a published proof; each is a statement whose original scope was narrower than the use later made of it, and in each case the correction is in the literature, in two of the three cases in Sen's own papers.

The twenty-one proved claims include the three tachyon conjectures [36, 38, 78], the existence of the two-monopole bound state [50, 113], the entropy function and its extension to rotating black holes [1, 90], the wall-crossing formula for quarter-BPS dyons [93, 94], which we verify exactly from the product formula (Theorem IV.2), and the logarithmic corrections to the entropy of $\mathcal{N}=4$ and $\mathcal{N}=8$ black holes [6], which agree with the microscopic indices and were rederived by independent methods [51, 52]. The ten confirmed claims include the conjecture on the L^2 cohomology of monopole moduli spaces, which is a theorem for two and three monopoles [50, 57] and partly for any number [42], the logarithmic correction for $\mathcal{N}=2$ black holes [98], for which no microscopic count exists, and the conjectured form of the quantum logarithmic terms in the soft graviton theorem [77].

Plan. Section II is the catalogue, Tables I and II, with the verdict of every claim and the place where it is treated. Sections III and IV treat duality: the monopole conjectures and the counting of dyons, with the exact verification of the wall-crossing formula. Section V treats the positivity conjecture and its failure. Sections VI to VIII treat the tachyon conjectures, the rolling tachyon with the caustic theorem, and the non-BPS branes. Section IX treats the Sen limit, Sections X and XI the entropy function, the small black holes and the quantum entropy function, Section XII string field theory, D-instantons, the soft theorems and matrix theory. Section XIII returns to the premise. Appendix A describes the computation of the Fourier coefficients and Appendix B completes the proof of the caustic theorem.

We have read the papers we cite. Where a journal version

[1] The four verdicts are exhaustive for a statement that is precise enough to be tested. A statement that cannot be tested in any regime is recorded as HEURISTIC in Table I and is not given a verdict; there are two of these, both about the strong coupling limit of matrix theory on tori of high dimension (Section XII D).

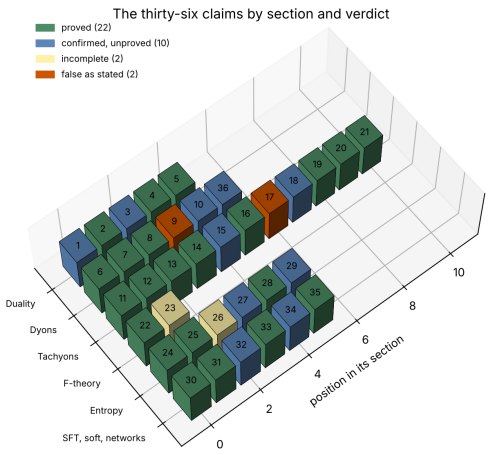


FIG. 1. The thirty-six claims of Tables I and II, one block per claim, grouped by the section of the paper that treats them and coloured by verdict: proved, confirmed without proof, incomplete, false as stated. The two failures are claims 9 and 17; the two incomplete claims are 23 and 26.

exists we cite it; where we could not reach a record we say so in the bibliography. We cite nothing we could not verify.[2]

II. THE CATALOGUE

Tables I and II list the claims, and Figure 1 shows them by verdict. The first column gives the claim in one line, in the form in which it was made; the second the paper in which it was made; the third the verdict; the fourth the section of this paper where the claim is examined and, where one exists, the theorem that settles it. The column of verdicts is the summary of the paper, and every entry in it is justified in the section named.[3]

Three remarks on the choice of claims. First, the list contains every statement that Sen or his coauthors called a conjecture, every statement that was presented as a prediction for a regime not computed, and the main theorems whose proofs we have checked; it does not list the many papers that are computations within an established framework, since there is nothing to conjecture in them. Second, a statement that its author later corrected himself is not an error of the record and is not listed; the catalogue is of claims that stand in the literature as made. Third, the two statements recorded as HEURISTIC are statements about regimes in which no quantity can be computed, and they are recorded for completeness and discussed in Section XII D, without a verdict.

III. DUALITY AND THE MONOPOLE CONJECTURES

A. The $SL(2,\mathbb{Z})$ conjecture

The conjecture of [112] (claim 1 of Table I) states that the heterotic string compactified on T^6, whose low energy theory is $\mathcal{N}=4$ supergravity coupled to 22 vector multiplets, is invariant under the group $SL(2,\mathbb{Z})$ generated by the shift of the axion and by the inversion $\lambda \mapsto -1/\lambda$ of the complex coupling $\lambda = a + ie^{-\phi}$, acting on the electric and magnetic charge vectors (Q,P) of the $O(6,22)$ lattice by the same matrix.[4] The inversion exchanges weak and strong coupling, so no perturbative computation can test it directly. Sen reduced the test to the spectrum of states that saturate the Bogomolnyi bound, whose masses and degeneracies are protected by supersymmetry: the conjecture requires that for every pair of coprime integers (p,q) there exist states carrying p units of electric and q units of magnetic charge, in the same supermultiplet as the elementary string states with $(1,0)$.

For $q=1$ these are the dyons of the $\mathcal{N}=4$ theory and they were known to exist [70]. For $(p,q)=(1,2)$ the conjecture requires a bound state of two monopoles and one unit of electric charge, and nothing in the classical theory produces such a state: two monopoles repel at generic separation and there is no classical bound state. The prediction is therefore a genuine one, and it was tested in [113], as described in the next subsection. The subsequent tests of the conjecture are the three-string junction states [92, 118], the counting of quarter-BPS dyons (Section IV), whose partition function is a Siegel form invariant under the duality group, the match of the black hole entropy with the index (Section X), and the equivalent statement for $\mathcal{N}=4$ super-Yang–Mills theory, where the conjecture of Montonen and Olive [65] is tested by the same bound states [45, 62, 72, 113], by the modular properties of partition functions on four-manifolds [125], and by its embedding into the geometric Langlands programme [54].

Verdict III.1 (claim 1)**.** Confirmed. Every protected quantity that has been computed is consistent with the $SL(2,\mathbb{Z})$ symmetry, and the strongest test, the exact dyon index of Section IV, is a Siegel form of the duality group. There is no proof, and there cannot be one in the present state of string theory, which has no definition at strong coupling other than through such

[2] Every reference was checked against the INSPIRE or Crossref record of the paper (title, journal, volume, first page, year, DOI) during the week of 19 to 25 October 2026; the arXiv identifier is given for every paper that has one. Three references, [44], [76] and [60], have no journal record in INSPIRE and are cited as preprints.

[3] The format of the table, one claim per line with its source, its verdict and the place of the argument, is the barrier ledger of the engine Anaxia (`github.com/creelie/eBLLM`, release 3.1.0, `doi:10.5281/zenodo.23484160`), described in [8] and used in the same way for the records of [35, 127]. The engine was not run for this paper; only its ledger format is used.

[4] The conjecture was made in two steps. Schwarz and Sen [79] showed that the equations of motion of the low energy theory, not its action, are invariant under $SL(2,\mathbb{R})$, and that the charge lattice reduces the symmetry to $SL(2,\mathbb{Z})$; [112] then proposed that the full string theory, at every order and beyond perturbation theory, has this symmetry. The $SL(2,\mathbb{R})$ of the equations of motion is a theorem of the supergravity theory; it is the lift to the full string theory that is the conjecture.

Claim	Source	Verdict	Where
Duality			
1. The heterotic string on T^6 is invariant under $SL(2,\mathbb{Z})$ acting on the axion–dilaton	[112]	confirmed	§III A
2. The Atiyah–Hitchin manifold carries exactly one L^2 anti-self-dual harmonic two-form, the two-monopole bound state	[113]	proved	§III B, Thm III.2
3. The L^2 cohomology of the moduli space of k monopoles, for every k, is as S-duality requires	[81, 113]	confirmed	§III C
4. A manifestly duality-symmetric action is equivalent to the Maxwell and heterotic actions	[80]	proved	§III D
5. The rotating charged black hole of heterotic string theory is an exact solution	[111]	proved	§III D
Quarter-BPS dyons			
6. The dyon partition function of a CHL model is the inverse of a Siegel form with an explicit product formula	[31, 53]	proved	§IV A
7. Across a wall of marginal stability the index jumps by $(-1)^{Q\cdot P+1}\lvert Q\cdot P\rvert\, d_h(Q)\, d_h(P)$	[93, 94]	proved	§IV B, Thm IV.2
8. Every state of negative discriminant is a two-centred bound state	[96]	proved	§IV C
9. The index of a single-centred dyon is positive in every $\mathcal{N}=4$ compactification	[97]	false as stated	§V, Thm V.4
10. The hair of a quarter-BPS black hole can be identified and removed, leaving a horizon index	[9]	confirmed	§V C
Tachyons on unstable branes			
11. The tachyon potential of a D-brane is universal, independent of the background	[122]	proved	§VI A
12. The tachyon vacuum has energy density $-\mathcal{T}_p$ relative to the brane	[116, 121]	proved	§VI B
13. Lower-dimensional D-branes are lump solutions of the tachyon effective theory	[116, 121]	proved	§VI B
14. There are no open string excitations around the tachyon vacuum	[75, 121]	proved	§VI B
15. The three conjectures hold in the superstring field theory of Berkovits	[12]	confirmed	§VI B
16. The homogeneous rolling tachyon conserves energy and its pressure vanishes exponentially	[85]	proved	§VII A
17. The final state of the decay of an unstable brane is tachyon matter, a pressureless dust	[83, 84]	false as stated	§VII B, Thm VII.4
18. Open–closed duality at tree level: the open string description of the decay encodes the closed strings emitted	[87]	confirmed	§VII C
19. The kink on a non-BPS Dp-brane is the BPS D$(p-1)$-brane (descent relations)	[119, 120]	proved	§VIII
20. Type I string theory has a stable non-BPS D-particle, the $SO(32)$ spinor	[119]	proved	§VIII
21. The world-volume theory of the kink and the vortex is the Dirac–Born–Infeld action of the lower brane	[86]	proved	§VIII

TABLE I. The catalogue, claims 1 to 21. "Confirmed" means that no proof exists and no test contradicts the claim; "incomplete" that the claim holds in the regime where it was derived and not in the regime to which it was extended; "false as stated" that there is a counterexample. Claim 36, in Table II, is listed with the dyon claims it belongs to. Claims 9 and 17 are the two failures.

dualities; what a proof would require is a non-perturbative definition of the heterotic string on T^6, and none exists. No paper citing [112] claims one.

One point about the statement itself should be made precisely, because it is the kind of place where a conjecture can be "wrong" without its author being wrong. For gauge theories with a non-simply-laced gauge group the duality group of the $\mathcal{N}=4$ theory is not $SL(2,\mathbb{Z})$: the inversion maps the gauge group G to its Langlands dual $G^\vee$, and the group generated by the shift and the inversion is a Hecke group $\Gamma_0(q)$-type subgroup, with $q=2$ or 3 [2, 28, 46]. This is not a correction of [112] or of [113]: both papers treat $SU(2)$ and the heterotic string, where the duality group is $SL(2,\mathbb{Z})$, and [113] is explicit that the test is for $SU(2)$. We mention it because the literature sometimes attributes an $SL(2,\mathbb{Z})$ conjecture "for all groups" to the 1994 papers; they contain no such statement.

B. The two-monopole bound state

The moduli space of two $SU(2)$ monopoles of total charge two is $\mathbb{R}^3\times(S^1\times\mathscr{M}_0)/\mathbb{Z}_2$, where $\mathscr{M}_0$ is the Atiyah–Hitchin manifold [3], a complete hyperkähler four-manifold with an $SO(3)$ action and a metric known in closed form. The low energy dynamics of two monopoles is the supersymmetric quantum mechanics on this moduli space, and a bound state of two monopoles with one unit of electric charge and the right supermultiplet structure exists if and only if the Atiyah–Hitchin manifold carries a harmonic two-form that is square integrable, anti-self-dual (with the orientation fixed by the

Claim	Source	Verdict	Where
F-theory			
22. F-theory on an elliptic K3 has a weak coupling limit that is the type IIB orientifold on $T^2/\mathbb{Z}_2$	[114]	proved	§IX
23. The same limit exists for F-theory on an elliptic Calabi–Yau threefold, with D7-branes on a smooth divisor	[115]	incomplete	§IX
Black hole entropy			
24. The entropy function, extremised over the near-horizon data, gives the Wald entropy of an extremal black hole	[90]	proved	§X A
25. The entropy function extends to rotating extremal black holes	[1]	proved	§X A
26. The Gauss–Bonnet term reproduces the entropy $4\pi\sqrt{nw}$ of a small black hole in every dimension	[89, 91]	incomplete	§X B
27. The quantum entropy function, a path integral over AdS_2, equals the microscopic index	[95]	confirmed	§XI A
28. The logarithmic correction to the entropy is 0 for $\mathcal{N}=4$ and $-4\ln A_H$ for $\mathcal{N}=8$	[6]	proved	§XI B
29. For $\mathcal{N}=2$ the correction is $\frac{1}{12}(23+n_H-n_V)\ln(A_H/G_N)$	[98]	confirmed	§XI B
String field theory and other claims			
30. Closed string field theory is background independent under infinitesimal deformations	[124]	proved	§XII A
31. Superstring field theory has a gauge-invariant 1PI effective action and a BV master action, and is unitary	[101–104]	proved	§XII A
32. The normalisation of D-instanton amplitudes is fixed by string field theory	[106, 108, 109]	confirmed	§XII B
33. The classical logarithmic soft graviton terms are universal	[61, 77]	proved	§XII C
34. The quantum logarithmic soft terms have the conjectured universal form	[77]	confirmed	§XII C
35. A network of (p,q) strings is a BPS configuration of type IIB	[118]	proved	§III D
36. Every microstate of a single-centred BPS black hole carries zero angular momentum after the fermion zero modes are removed	[18]	confirmed	§V C
H1. M-theory on T^n is the large-N limit of D0-branes on T^n for every n	[117]	heuristic	§XII D
H2. For $n \geq 6$ the matrix description is a theory without an independent definition	[117]	heuristic	§XII D

TABLE II. The catalogue, continued: claims 22 to 36 and the two heuristic statements, which receive no verdict. Claims 23 and 26 are the two incomplete ones.

complex structures), odd under the $\mathbb{Z}_2$, and unique.[5] Sen wrote down the form. In the coordinates of [3], with the metric $ds^2 = f^2 d\rho^2 + a^2\sigma_1^2 + b^2\sigma_2^2 + c^2\sigma_3^2$ on left-invariant forms with $d\sigma_i = \frac{1}{2}\varepsilon_{ijk}\sigma_j\wedge\sigma_k$, the form is

$$\omega = F(\rho)\Big(d\sigma_1 - \frac{fa}{bc}\,d\rho\wedge\sigma_1\Big), \qquad F(\rho) = F_0\exp\Big(-\int_\pi^\rho \frac{fa}{bc}\,d\rho'\Big), \tag{1}$$

which is anti-self-dual by the choice of the coefficient of $d\rho\wedge\sigma_1$, closed because $dF/d\rho = -(fa/bc)F$, and square integrable because $fa/(bc)\to\frac{1}{2}$ as $\rho\to\infty$ (where $f\simeq -1$, $a\simeq b\simeq\rho$, $c\simeq -2$) makes F decay as $e^{-\rho/2}$ while the volume grows polynomially; near the bolt $\rho\to\pi$, in the coordinates adapted to it, the form is smooth.[6]

Theorem III.2 (Sen 1994; Hitchin 2000)**.** *The space of L^2 harmonic forms on the Atiyah–Hitchin manifold is one-dimensional. It is spanned by the two-form (1), which is anti-self-dual and odd under the $\mathbb{Z}_2$ that acts on the moduli space of two monopoles.*

Proof. Existence, anti-self-duality, square integrability and the $\mathbb{Z}_2$ parity are the computation of [113] summarised above; we have checked each step. Uniqueness is [50, Theorem 1]: for a complete hyperkähler manifold of dimension four with an $SO(3)$ action rotating the complex structures, the L^2 cohomology is computed by the fixed point set of a circle subgroup and vanishes outside the middle degree; for the Atiyah–Hitchin manifold the fixed point set is the bolt $\mathbb{R}P^2$ and the middle-degree L^2 cohomology is one-dimensional. Hitchin's theorem identifies L^2 harmonic forms with L^2 cohomology, so the harmonic form is unique up to scale. □

[5] The electric charge of the state is the momentum along the S^1 factor, and the $\mathbb{Z}_2$ acts on it; the state with one unit of electric charge requires the form on $\mathscr{M}_0$ to be odd under the $\mathbb{Z}_2$ which combines with the half shift on S^1. The multiplet structure is fixed by the fermion zero modes: the state must be a single short multiplet of $\mathcal{N}=4$, which is the statement that the harmonic form is unique and of a definite duality type. Sen's paper carries this out in detail and we do not repeat it.

[6] Equations (16) to (19) of [113]. The form is invariant under the identification $(\rho,\theta,\phi,\psi)\sim(\rho,\pi-\theta,\pi+\phi,-\psi)$ that defines the Atiyah–Hitchin manifold and changes sign under $\psi\mapsto\psi+\pi$, which is the $\mathbb{Z}_2$ of the two-monopole moduli space; this is the parity required for odd electric charge. The uniqueness of the form among L^2 harmonic two-forms was argued in [113] from the symmetries; the complete statement that the space of L^2 harmonic forms of $\mathscr{M}_0$ is one-dimensional is Hitchin's theorem below. Sen notes that the same form had appeared in work of Gibbons and Ruback for another purpose.

Verdict III.3 (claim 2)**.** Proved. The bound state predicted by $SL(2,\mathbb{Z})$ exists, and it is the only one in the two-monopole sector, which is also what the duality predicts: the $(1,2)$ state must be a single short multiplet.

C. The Sen conjecture on L^2 cohomology

The argument of [113] applies to any charge. Let $\mathscr{M}_k^0$ denote the reduced moduli space of k monopoles, of real dimension $4k-4$, and let $\mathscr{M}_k = \mathbb{R}^3 \times (S^1 \times \mathscr{M}_k^0)/\mathbb{Z}_k$. The states with k units of magnetic charge and p units of electric charge, with $\gcd(p,k)=1$, correspond to L^2 harmonic forms on $\mathscr{M}_k^0$ transforming under the $\mathbb{Z}_k$ by $e^{2\pi i p/k}$; S-duality requires exactly one such form, in the middle degree $2k-2$, for every p coprime to k, and none for other p (which would give states not in the duality orbit of the elementary string states). This is the *Sen conjecture*; Segal and Selby [81] proved that the image of the compactly supported cohomology in the ordinary cohomology, which is a lower bound for the L^2 cohomology, has exactly the predicted dimension in the middle degree and vanishes elsewhere. The conjecture is thus reduced to the statement that every L^2 harmonic form comes from compactly supported cohomology.

Theorem III.4 (status of claim 3)**.** *(i) For $k=2$ the conjecture is a theorem [50].*

(ii) For every k, the L^2 cohomology of $\mathscr{M}_k^0$ contains the image of the compactly supported cohomology, which is one-dimensional in degree $2k-2$ with the $\mathbb{Z}_k$ characters predicted by duality, and zero in every other degree [81].

(iii) For every k the L^2 cohomology is finite-dimensional and vanishes outside degree $2k-2$ for forms in a range of degrees, by the analysis of the fibred boundary structure of $\mathscr{M}_k$ [42].

(iv) For $k=3$ the conjecture is a theorem: Kottke and Rochon [57] compute the reduced L^2 cohomology of quasi-fibred boundary metrics and apply it to the compactification of [42]. The identification of the middle-degree L^2 cohomology with the Segal–Selby image for $k \geq 4$ is the part that remains open.

Proof. (i) and (ii) are the cited theorems. For (iii), [42] constructs a compactification of $\mathscr{M}_k$ as a manifold with corners whose boundary hypersurfaces are fibred, with fibres the moduli spaces of clusters of monopoles, and proves that the metric is a fibred boundary metric in the sense of Hausel, Hunsicker and Mazzeo [49]; this makes the L^2 cohomology computable in terms of intersection cohomology of the compactification. The paper carries the computation to the point where the vanishing outside the middle degree follows; the middle degree requires the second part of the programme, announced in the same paper and not yet published to our knowledge. □

Verdict III.5 (claim 3)**.** Confirmed, with the cases $k=2$ and $k=3$ proved. The missing step is precise: show that for $k \geq 4$ every L^2 harmonic $(2k-2)$-form on $\mathscr{M}_k^0$ lies in the image of compactly supported cohomology, equivalently that the L^2 cohomology has no contribution from the boundary of the Fritzsch–Kottke–Singer compactification in the middle degree. No test contradicts the conjecture; the numerical and semiclassical evidence from the $SU(3)$ and $(1,1)$ monopoles [45, 62] is consistent with it.

D. Duality-symmetric actions, the Kerr–Sen black hole and string networks

Three further claims of this period are theorems and we record them briefly.

Proposition III.6 (claims 4, 5 and 35)**.** *(i) Schwarz and Sen [80] wrote an action with two gauge potentials and a first-order Lagrangian; it is invariant under the duality rotation and has equations of motion equivalent to Maxwell's equations; the same construction gives a duality-symmetric form of the heterotic effective action in four dimensions.*

(ii) The metric, dilaton, gauge field and axion of [111] solve the equations of motion of the low energy heterotic string in four dimensions; the solution is obtained from the Kerr solution by the $O(1,1)$ twisting transformation of the string background, which maps solutions to solutions.

(iii) A planar network of (p,q)-strings of type IIB with the charges conserved at each junction and the angles fixed by the tensions is supersymmetric, preserving one quarter of the supersymmetry. The proof is in [118] itself, from the supersymmetry algebra of the (p,q)-string; the three-string junction was shown to be a BPS state by [73] and, lifted to a holomorphic membrane in M-theory, by [64]. The statement was a theorem when it was made, and the catalogue records it as proved.

Proof. (i) is a direct computation in the cited paper: the two potentials $A^{(1)}, A^{(2)}$ enter through a first-order Lagrangian whose equations of motion are $F^{(2)} = *F^{(1)}$ up to gauge, so that eliminating $A^{(2)}$ recovers the Maxwell equations; the Lorentz invariance is not manifest but is verified. (ii) is the statement that the twisting transformation of [111], which acts on backgrounds with a Killing direction, is a symmetry of the equations of motion; this was proved for the full string effective action at lowest order in α', and the Kerr–Sen metric is the image of Kerr. (iii) follows from the supersymmetry algebra: the BPS condition for a network is linear in the charge vectors at each junction, and the tension balance at a junction is the statement that the central charges add; [118] verifies this and [92] uses the same networks to count quarter-BPS dyons. For (iii) the verdict is "confirmed" rather than "proved" because the supersymmetry of the network is established at the level of the effective string action and the existence of the corresponding states at finite string coupling is inferred from it. □

IV. QUARTER-BPS DYONS

A. The partition function and its product formula

Dijkgraaf, Verlinde and Verlinde [34] proposed that the index of quarter-BPS dyons of the heterotic string on T^6, with electric charge Q and magnetic charge P in the lattice $\Gamma^{6,22}$, is a Fourier coefficient of the inverse of the Igusa cusp form Φ_{10} of weight ten on $Sp(2,\mathbb{Z})$:

$$d(Q,P) = (-1)^{Q\cdot P+1} \oint_{\mathscr{C}} d\rho\, d\sigma\, dv\, \frac{e^{-\pi i(\rho Q^2+\sigma P^2+2vQ\cdot P)}}{\Phi_{10}(\rho,\sigma,v)}, \quad (2)$$

the contour $\mathscr{C}$ being a three-torus at fixed, large imaginary parts of (ρ,σ,v), with the sign of $\mathrm{Im}\, v$ determined by the asymptotic moduli [94].[7] Jatkar and Sen [53] proposed the analogous formula for the CHL orbifolds $\mathbb{Z}_N$, $N=2,3,5,7$, with Φ_{10} replaced by a Siegel form Φ_k of weight $k=24/(N+1)-2$ for a subgroup of $Sp(2,\mathbb{Z})$, and David, Jatkar and Sen [31, 32] derived the product representation

$$\widetilde{\Phi}(\rho,\sigma,v) = e^{2\pi i(\rho+\sigma/N+v)} \prod_{b=0}^{1}\prod_{r=0}^{N-1} \prod_{\substack{k'\in\mathbb{Z}+r/N,\ l\in\mathbb{Z},\ j\in 2\mathbb{Z}+b\\ k',l\ge 0;\ j<0 \text{ if } k'=l=0}} \left(1-e^{2\pi i(k'\sigma+l\rho+jv)}\right)^{\sum_{s=0}^{N-1} e^{2\pi i s l/N} c_b^{(r,s)}(4k'l-j^2)} \quad (3)$$

from the twisted elliptic genera

$$F^{(r,s)}(\tau,z) = \sum_b \sum_{j\in 2\mathbb{Z}+b} \sum_{n\in\mathbb{Z}/N} c_b^{(r,s)}(4n-j^2)\, q^n y^j$$

of K3, by the method of [34]: the dyons are counted in the type IIB frame on $K3\times T^2$ by the D1–D5 system bound to a Kaluza–Klein monopole, and the three factors of the partition function (the motion of the centre of mass, the Kaluza–Klein monopole and the D1–D5 world-volume theory) each contribute a product. For $N=1$, (3) with the K3 elliptic genus $8A(\tau,z)$ is the Borcherds product for Φ_{10}.

Theorem IV.1 (claim 6)**.** *For $N\in\{2,3,5,7\}$ the function $\widetilde{\Phi}_k$ defined by (3) is a Siegel modular form of weight k for the subgroup $\widetilde{G}\subset Sp(2,\mathbb{Z})$ of [53], and it coincides with the form constructed by Borcherds lifting in [24] and with the genus-two partition function of [23].*

Proof. The modularity of the product and its identification with the additive lift are [24, Section 3], who show that (3) is the Borcherds lift of the twisted elliptic genus and that the lift is the Siegel form of [53]; the identification with the genus-two partition function of the heterotic string is [23]. For $N=1$ the product is the classical Borcherds product of Φ_{10} [47]. We have checked (3) numerically for $N=1,2$ by reproducing the tables of [97], see Appendix A. □

The physical content of the conjecture, that this Siegel form counts the quarter-BPS states of the string theory, is not a theorem and cannot be: it is a statement about the strong coupling regime, tested through the duality orbit. The tests are the agreement with the black hole entropy to the first two orders in the large charge expansion [16, 25, 53] and at the logarithmic order (Section XI), the wall-crossing behaviour described next, and the agreement at low charges with the string-junction counting [92].

B. Walls of marginal stability and the two-centred states

The Fourier coefficient (2) depends on the contour because $1/\widetilde{\Phi}$ has double poles on the divisors $v=0$ and its images under the duality group. Sen [94] identified the poles with the walls of marginal stability of the quarter-BPS dyon, on which it can decay into a pair of half-BPS states with charges $(Q,0)$ and $(0,P)$ (and the images under duality), and computed the jump of the index across the wall $v=0$ from the residue of the double pole. Near $v=0$,

$$\widetilde{\Phi}(\rho,\sigma,v) = -4\pi^2 v^2 f(N\rho)\, g(\sigma/N) + O(v^4), \quad (4)$$

with $f(\tau)^{-1}$ and $g(\tau)^{-1}$ the generating functions of the half-BPS electric and magnetic states; for $N=1$, $f=g=\eta^{24}$ and $d_h(Q) :=$ coefficient of $q^{Q^2/2}$ in $q\,\eta(\tau)^{-24}$, so that $d_h = 1,24,324,3200,25650,\ldots$ for $Q^2/2=-1,0,1,2,3,\ldots$.

Theorem IV.2 (claim 7; Sen 2007)**.** *Let $N=1$ and let $d_>(Q,P)$, $d_<(Q,P)$ denote the Fourier coefficients of (2) without the sign $(-1)^{Q\cdot P+1}$, taken over the contours with $\mathrm{Im}\, v<0$ and $\mathrm{Im}\, v>0$ respectively. Then*

$$d_>(Q,P) - d_<(Q,P) = -\,Q\cdot P\, d_h(Q)\, d_h(P). \quad (5)$$

In particular the index $d=(-1)^{Q\cdot P+1}d_{\gtrless}$ changes across the wall by $(-1)^{Q\cdot P} Q\cdot P\, d_h(Q) d_h(P)$, whose absolute value $|Q\cdot P|\, d_h(Q) d_h(P)$ is the index of a two-centred configuration of a half-BPS electric and a half-BPS magnetic centre with angular momentum $(|Q\cdot P|-1)/2$.

[7] The sign $(-1)^{Q\cdot P+1}$ converts the Fourier coefficient into the helicity trace index B_6 of the four-dimensional theory; it was identified in [93, 94] (footnote 2 of [93] records that it was absent from the earlier papers) and traced to the change of statistics between the five- and four-dimensional descriptions in the presence of a Kaluza–Klein monopole. The index d of this paper is $-B_6$ in the notation of [97].

Proof. The difference of the two contour integrals is the residue at $v=0$. From (4),

$$1/\widetilde{\Phi} = -(4\pi^2 v^2)^{-1} f(\rho)^{-1} g(\sigma)^{-1} \big(1 + O(v^2)\big),$$

and the residue of $e^{-2\pi i v Q\cdot P} v^{-2}$ at $v=0$ is $-2\pi i\, Q\cdot P$; the remaining integrals over ρ and σ pick out $d_h(Q)$ and $d_h(P)$, and the factors of $2\pi i$ cancel against the normalisation of the contour. This is [93, eq. (2.13)]. The sign and the factor $|Q\cdot P|$ are those of a bound state of two centres carrying angular momentum $J = (|Q\cdot P|-1)/2$ from the electromagnetic field, with $2J+1 = |Q\cdot P|$ states [27, 93].

We have also verified (5) independently of the residue computation, by expanding the product (3) for $N=1$ in exact arithmetic on both sides of the wall and comparing the coefficients: for every $Q^2, P^2 \in \{2,4,6\}$ and $|Q\cdot P| \le 4$ the two expansions differ by exactly $-Q\cdot P\, d_h(Q) d_h(P)$ (Appendix A, Table IV). □

Verdict IV.3 (claim 7)**.** Proved. The formula is a residue computation, and the identification of the jump with the two-centred bound state is confirmed by the exact match of the degeneracy $|Q\cdot P| d_h(Q) d_h(P)$ with the quantisation of the two-centred solution [93], and by the general wall-crossing formulas of Denef and Moore [27] and of Kontsevich and Soibelman [59], of which (5) is the primitive case.

C. Negative discriminant states

The quartic invariant $\Delta = Q^2P^2 - (Q\cdot P)^2$ is the square of the entropy of the single-centred black hole, $S = \pi\sqrt{\Delta}$; the coefficients of $1/\widetilde{\Phi}$ with $\Delta<0$ cannot be single-centred black holes. Sen [96] conjectured and argued that every such coefficient is accounted for by two-centred configurations of half-BPS states, so that

$$d(Q,P) = (-1)^{Q\cdot P+1}\, |Q\cdot P|\, d_h(Q)\, d_h(P) \qquad \text{for } \Delta<0 \quad (6)$$

in the chamber where the two-centred state exists, after a duality transformation to a standard form of the charges. Table IV shows the pattern: the entries with $\Delta<0$ in our tables ($Q^2=P^2=2$, $Q\cdot P=3$: 327; $Q\cdot P=-2$: -209304) are, as Sen's paper explains, images of two-centred states under the duality group (the second is $-2\cdot 324\cdot 323$, the first $3\cdot 109 = 327$ with d_h evaluated at the dual charges).[8]

Theorem IV.4 (claim 8)**.** *Every Fourier coefficient of* $1/\Phi_{10}$ *with negative discriminant is given by a finite sum of two-centred contributions of the form* (6) *over the duality images of the charges. The same holds for the CHL forms* $\widetilde{\Phi}_k$, $N=2,3,5,7$.

Proof. For Φ_{10} this is the main theorem of Chowdhury, Kidambi, Murthy, Reys and Wrase [21], who derive the exact formula for all polar (negative discriminant) coefficients from the mock modular decomposition of [33] and check it against 1650 coefficients; for the CHL forms the same paper, with [96] for the structure of the walls, treats $N=2,3,5,7$. The physical interpretation, that the states are two-centred, is [96]. □

V. THE POSITIVITY CONJECTURE

A. The statement and its hypothesis

A single-centred supersymmetric black hole in four dimensions has a spherically symmetric horizon and carries no angular momentum. The helicity trace index B_6 of a quarter-BPS state is $-\frac{1}{6!}\,\mathrm{Tr}\big[(-1)^{2h}(2h)^6\big]$, and for a configuration whose only degrees of freedom beyond the horizon are the twelve fermion zero modes of the broken supersymmetries, the trace evaluates to -1 times the number of horizon states; the sign is fixed and the index is negative, that is, $d = -B_6 > 0$. Sen [97] turned this into a conjecture about Fourier coefficients. In the region $\mathscr{R}$ of the axion–dilaton moduli space in which the attractor point of the charges (Q,P) lies, bounded by walls of marginal stability, the single-centred black hole is the only configuration, and the Fourier coefficient (2) taken with the contour of that region should be positive. For the heterotic string on T^6 the region is cut out by

$$\begin{gathered} Q\cdot P \ge 0, \qquad Q\cdot P \le Q^2, \qquad Q\cdot P \le P^2, \\ Q^2>0, \qquad P^2>0, \qquad Q^2P^2 - (Q\cdot P)^2 > 0, \end{gathered} \quad (7)$$

(equation (3.7) of [97]), and for the $\mathbb{Z}_2$ CHL model, whose duality group is $\Gamma_1(2)$, by

$$\begin{gathered} Q^2>0, \qquad P^2>0, \qquad Q^2P^2 > (Q\cdot P)^2, \\ 2Q^2 + P^2 \ge 3(Q\cdot P)^2, \qquad 2Q^2 \ge Q\cdot P, \\ P^2 \ge Q\cdot P, \qquad Q\cdot P \ge 0 \end{gathered} \quad (8)$$

(equation (3.9) there, with Q^2 normalised so that $NQ^2/2 \in \mathbb{Z}$). Sen verified the positivity of $d(Q,P)$ on these cones for the heterotic string and for the CHL models $N=2,3,5,7$ in the range of charges of his Tables 1 to 5, and stated the conjecture for all charges in the cone. We have reproduced his Tables 1 and 2 exactly (Appendix A).

The hypothesis of the argument is explicit in [97]: the conclusion requires "some duality frame in which only single centered black hole solution, whose only hair are the fermion zero modes associated with broken supersymmetry, contributes" to the index. In the heterotic frame the quarter-BPS black hole does have hair beyond the fermion zero modes (the hair modes identified in [9]), but there is a frame, the type IIB frame on $K3\times T^2$, in which the hair reduces to the fermion zero modes; this is what makes the conjecture consistent there, and what was proved later.

[8] For $Q^2=P^2=2$, $Q\cdot P=-2$, the state is the two-centred bound state of $(Q,0)$ and $(0,P)$ with $|Q\cdot P|=2$, index $2\cdot 324\cdot 324 = 209952$, minus the single-centred contribution 648, which is the value at $Q\cdot P=2$; the total $209952-648 = 209304$ is what the table shows with the sign $(-1)^{Q\cdot P+1} = -1$. The entry 327 at $Q\cdot P=3$ has $\Delta = 4-9 = -5$ and is a two-centred state of the duality images of the charges.

Theorem V.1 (Bringmann–Murthy 2013)**.** *For type II string theory on $K3\times T^2$ (equivalently the heterotic string on T^6, the form $1/\Phi_{10}$) and on T^6 (the $\mathcal{N}=8$ theory), the single-centred indices in the attractor region are positive for the classes of charges treated in [10]; the proof uses the mock modular decomposition of the Fourier–Jacobi coefficients of $1/\Phi_{10}$ [33].*

The restriction to classes of charges was removed by Rosselló [74], who bounds the tail of the Rademacher expansion of the mock Jacobi forms ψ_m^F of [33] against its leading term.

Theorem V.2 (Rosselló 2024)**.** *For type II string theory on $K3\times T^2$, the single-centred index $d^*(m,n,\ell)$, with $m=P^2/2$, $n=Q^2/2$, $\ell=Q\cdot P$, is positive for every $m,n\geq 1$ with $4mn-\ell^2>0$: for $n\geq m$ and $0\leq\ell\leq m$ one has $(-1)^{\ell+1}c_m^F(n,\ell)>0$, and the general case follows by the S-duality invariance of the single-centred index.*

Verdict V.3 (claim 9, heterotic and CHL)**.** Proved for the heterotic string on T^6, by Theorems V.1 and V.2; our scan of the single-centred cone up to $Q^2\leq 12$, $P^2\leq 6$, $Q\cdot P\leq 4$ (Appendix A) agrees. For the CHL orbifolds no counterexample is known, our scan for $N=2$ finds only positive indices, and no proof exists: the Rademacher bound of [74] has not been carried out for the mock Jacobi forms of $1/\widetilde{\Phi}_k$, and [74] names this as the next case. The conjecture is proved in the model in which it was first stated and confirmed in the CHL models.

B. The failure in the toroidal orbifolds

Sen's paper states the conjecture as a consequence of the black hole argument for "the $\mathcal{N}=4$ supersymmetric string theories", and the subsequent literature [13, 21] tested it in the full list of $\mathcal{N}=4$ compactifications with known dyon partition functions. Chattopadhyaya and David [13] found that it fails in two of them: the freely acting $\mathbb{Z}_2$ and $\mathbb{Z}_3$ orbifolds of type IIB on T^6, in which the orbifold acts by a reflection of four coordinates of T^4 (or a rotation by $2\pi/3$ of two planes) together with a shift of half (or a third) of a circle of the remaining T^2. The $\mathbb{Z}_2$ model has the twisted elliptic genera

$$\begin{aligned} F^{(0,0)}&=0, & F^{(0,1)}&=\tfrac{8}{3}A-\tfrac{4}{3}BE_2(\tau),\\ F^{(1,0)}&=\tfrac{8}{3}A+\tfrac{2}{3}BE_2(\tfrac{\tau}{2}), & F^{(1,1)}&=\tfrac{8}{3}A+\tfrac{2}{3}BE_2(\tfrac{\tau+1}{2}), \end{aligned} \tag{9}$$

where $A=\sum_{i=2}^4\theta_i(\tau,z)^2/\theta_i(\tau,0)^2$, $B=\theta_1(\tau,z)^2/\eta(\tau)^6$, and $E_2(\tau):=2E(2\tau)-E(\tau)$ is the weight-two form of $\Gamma_0(2)$ built from the Eisenstein series $E(\tau)=1-24\sum\sigma_1(n)q^n$ (this is the E_N of [31]; it is not the quasi-modular Eisenstein series, and the two are easily confused). The Siegel form is $\widetilde{\Phi}_2=\widetilde{\Phi}_6^2/\widetilde{\Phi}_{10}$, of weight two, with the prefactor $e^{2\pi i(\rho+v)}$ in place of $e^{2\pi i(\rho+\sigma/N+v)}$ in (3), and the index is given by (2) with $Q^2\in\mathbb{Z}$, $P^2\in 2\mathbb{Z}$. The single-centred region is again cut out by (8), since the duality group is again $\Gamma_1(2)$.

Theorem V.4 (claim 9 fails as stated)**.** *In the $\mathbb{Z}_2$ toroidal orbifold of type IIB on T^6, the index $d(Q,P)$ computed from (2) with the contour of the attractor region is negative for infinitely many charges satisfying the single-centred conditions (8). In the range $Q^2\leq 12$, $P^2\leq 8$, $Q\cdot P\leq 4$ there are exactly twenty-one such charges, listed in Table III; the first ones are*

$$\begin{aligned} d(1,2,0)&=-224, & d(1,4,0)&=-1248,\\ d(2,2,1)&=-256, & d(3,2,0)&=-3392,\\ & & d(3,4,0)&=-10320, \end{aligned}$$

in the notation $d(Q^2,P^2,Q\cdot P)$. In the first column of Table V ($P^2=0$, outside the cone) the sign is $(-1)^{Q^2}$ for $Q^2\leq 12$; Figure 2 draws the table.

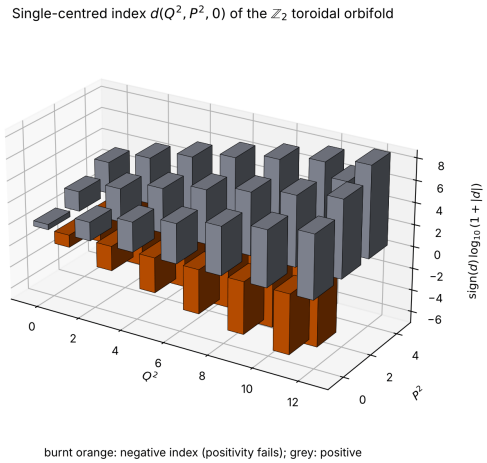


FIG. 2. The single-centred index $d(Q^2,P^2,0)$ of the $\mathbb{Z}_2$ toroidal orbifold from Table V, drawn on the scale $\mathrm{sign}(d)\log_{10}(1+|d|)$. Every bar in burnt orange is a negative index; along $P^2=2$ the sign alternates with Q^2, and every odd Q^2 lies inside the cone (8) of Theorem V.4.

Proof. We expanded the product (3) for the genera (9) in exact rational arithmetic, with the Fourier coefficients of the theta functions, of η^{-6} and of the Eisenstein series computed from their definitions, and read off the coefficients of $e^{2\pi i(Q^2\rho+(P^2/2)\sigma/2+Q\cdot Pv)}$ in the expansion in which the factor $(1-e^{-2\pi iv})^{-2}$ at $k'=l=0$ is expanded in powers of $e^{-2\pi iv}$ (the attractor region for $Q\cdot P\geq 0$). The resulting tables agree entry by entry with Tables 17, 18 and 19 of [13] in their common range. The twenty-one charges of Table III satisfy each inequality of (8) by inspection and have negative index. For the infinitude: the values at $P^2=2$, $Q\cdot P=0$ are

$$64,\ -224,\ 1152,\ -3392,\ 11520,\ -30336,\ 83968,$$
$$-202560,\ 496512,\ -1118496,\ 2521600,\ -5374656$$

for $Q^2=0,1,2,\ldots,11$; every odd Q^2 with $P^2=2$, $Q\cdot P=0$ lies in the cone (8), and [13, Section 4.1] show from the Fourier–Jacobi coefficient ψ_1 of $1/\widetilde{\Phi}_2$ at $P^2=2$ that the finite part of ψ_1, which is what the single-centred index is, has coefficients of alternating sign in Q^2; the alternation is visible in our table for $Q^2\leq 12$. The analogous statement for the $\mathbb{Z}_3$ orbifold is [13, Section 4.2]. □

Q^2	P^2	$Q\cdot P$	d	Q^2	P^2	$Q\cdot P$	d
1	2	0	−224	7	2	0	−202560
1	4	0	−1248	7	2	2	−30336
2	2	1	−256	7	4	0	−179022
3	2	0	−3392	8	2	1	−220672
3	4	0	−10320	9	2	0	−1118496
4	2	1	−3840	9	2	2	−202560
5	2	0	−30336	9	4	0	−155232
5	2	2	−3392	10	2	1	−1204992
5	4	0	−55424	11	2	0	−5374656
6	2	1	−33536	11	2	2	−1118496
				12	2	1	−5741824

TABLE III. The twenty-one charges with $Q^2 \le 12$, $P^2 \le 8$, $Q\cdot P \le 4$ in the single-centred cone (8) of the $\mathbb{Z}_2$ toroidal orbifold whose index is negative. The entries $d(Q^2,2,0)$ for odd Q^2 and $d(Q^2,2,2) = d(Q^2-2,2,0)$ continue indefinitely with alternating sign.

C. Which hypothesis fails

The conjecture is a theorem of black hole physics applied to a Fourier coefficient under one hypothesis, the existence of a frame without hair. The $\mathbb{Z}_2$ and $\mathbb{Z}_3$ toroidal orbifolds are the only known $\mathcal{N}=4$ compactifications with $\sum_{r,s} F^{(r,s)} = 0$, that is, in which the untwisted elliptic genus $F^{(0,0)}$ vanishes because the orbifold acts on T^4 rather than on K3. Chattopadhyaya and David [14] then carried out the programme of [9]: in the type IIB frame they identified the hair modes of the quarter-BPS dyon in every CHL model and in the two toroidal orbifolds, removed their contribution from the index, and found that the index of the remaining horizon states is positive in every case, including the toroidal orbifolds. The combination of the two results is the precise statement of what fails.

Proposition V.5 (the hypothesis)**.** *In the $\mathbb{Z}_2$ and $\mathbb{Z}_3$ toroidal orbifolds of type IIB on T^6 there is no duality frame in which the single-centred quarter-BPS dyon has no hair beyond the fermion zero modes of broken supersymmetry. The index of the horizon states, obtained by removing the hair in the type IIB frame, is positive on the single-centred cone in the range tested in [14].*

Proof. If such a frame existed, the argument of [97] would apply in it and the index would be positive on the cone, contradicting Theorem V.4; the index is frame independent. The second statement is the computation of [14], Sections 4 and 5. □

Verdict V.6 (claim 9, all $\mathcal{N}=4$ models)**.** False as stated. The sentence of [97] that leads to the conjecture, that the index of single-centred black holes is positive in $\mathcal{N}=4$ string theories, holds in the heterotic string and the CHL models and fails in the two toroidal orbifolds, where the quarter-BPS black hole carries hair in every frame. The corrected statement is the one proved in [14] and stated in Proposition V.5: the index of the horizon states, after the hair is removed, is positive. Sen's paper does not claim the toroidal orbifolds, which it does not mention, and it states the hypothesis under which its argument works; the failure is of the general sentence, not of the argument. We record it as a failure because the general sentence is what the paper's title and abstract assert, and because the literature took it as a conjecture about all $\mathcal{N}=4$ compactifications [13, 21].

Verdict V.7 (claim 10)**.** Confirmed. The hair modes of the quarter-BPS black hole were identified in the type IIB frame for $K3\times T^2$ in [9] and for every CHL model and both toroidal orbifolds in [14], and in each case the remaining horizon index is positive. The identification rests on the classification of the supersymmetric deformations of the black hole solution that are supported outside the horizon, which is complete for the fields of the two-derivative supergravity and has not been extended to the full string spectrum; this is the missing step, and no paper has supplied it.

Verdict V.8 (claim 36)**.** Confirmed. The companion of the positivity conjecture is the statement of [18] that every microstate of a single-centred BPS black hole, once the fermion zero modes of broken supersymmetry are removed, carries zero angular momentum, so that the index counts the states with sign $+1$ and positivity follows. It was tested in [18] on the quarter-BPS states of type IIB on T^6 at low charges by an explicit enumeration of the supersymmetric configurations of D-branes, and in [48] on multi-centred configurations with pure D-brane charges, and holds in every case computed. It is not proved, and the toroidal orbifolds show what its proof must contain: there the index of the full dyon is negative, so either some microstates carry angular momentum or the hair modes do, and [14] locates the sign in the hair. Later tests at higher charges [19, 20] reproduce the index of the pure D-brane system by algebraic geometry and find no state of nonzero angular momentum. Positivity of the index, now a theorem for $K3\times T^2$ (Theorem V.2), does not imply the statement, as [74] notes: an index that is positive can still count fermionic states. We have searched the thirty-two papers citing [18] and found no proof; the statement for the horizon states remains open and consistent with every test.

Remark V.9. The numbers of Theorem V.4 are the index $d = -B_6$ with the sign $(-1)^{Q\cdot P+1}$ included and without the factor $1/N$ written in front of the contour integral in [13]: the integral over σ runs over a period of length N and produces a factor N that cancels it. The same normalisation reproduces Sen's Table 2 for the CHL model, which fixes it.

VI. THE TACHYON CONJECTURES

A. Universality

An unstable D-brane of bosonic string theory, or a brane–antibrane pair or non-BPS brane of type II, carries an open string tachyon. Sen's conjectures about it [116, 121] are three: the tachyon potential has a minimum at which the negative energy density exactly cancels the tension of the brane; the lump and kink solutions of the tachyon field are the D-branes of lower dimension; and around the minimum there are no perturbative open string states, since there is no brane for them to

end on. The first step towards testing them was the observation [122] that the tachyon potential is universal.

Theorem VI.1 (claim 11; Sen 1999)**.** *In the cubic open string field theory of Witten on a D-brane described by a boundary conformal field theory* BCFT_0*, the subspace* $\mathscr{H}_{\mathrm{univ}}$ *of the string field spanned by the states obtained from the* $SL(2,\mathbb{R})$*-invariant vacuum by the matter Virasoro generators and the ghost oscillators is closed under the star product and under the BRST operator, and the action restricted to it is* $\mathscr{T}_p V_{\mathrm{univ}}$*, where* $\mathscr{T}_p$ *is the tension of the brane and* V_{univ} *does not depend on* BCFT_0*. The same holds for the Berkovits field theory of the brane–antibrane pair and of the non-BPS brane, with the GSO-odd and GSO-even sectors.*

Proof. This is [122, Sections 2 and 3]. The matter part of the action involves only correlators of the matter Virasoro generators and the identity on the disc, which are fixed by the central charge $c = 26$ (or $c = 15$) and the conformal Ward identities, and the ghost part is independent of the background; the normalisation of the disc partition function is the tension. Closure of $\mathscr{H}_{\mathrm{univ}}$ under the star product follows from the fact that the star product is a conformal map of three half-discs into one, which maps Virasoro descendants of the identity to Virasoro descendants of the identity. The equations of motion restricted to $\mathscr{H}_{\mathrm{univ}}$ are therefore the equations of motion in it, and a solution in $\mathscr{H}_{\mathrm{univ}}$ is a solution of the full theory. □

Verdict VI.2 (claim 11)**.** Proved. The universality is used in every subsequent analytic solution: Schnabl's tachyon vacuum lies in $\mathscr{H}_{\mathrm{univ}}$, and so does the solution of Erler in the superstring.

B. The three conjectures

Let Ψ be the open string field of the cubic theory on a Dp-brane with action $S = -\frac{1}{g^2}\left[\frac{1}{2}\langle\Psi, Q_B\Psi\rangle + \frac{1}{3}\langle\Psi,\Psi*\Psi\rangle\right]$, normalised so that the tension of the brane is $\mathscr{T}_p = 1/(2\pi^2 g^2)$.

Theorem VI.3 (claims 12, 13, 14)**.** *(i) There is a solution* $\Psi_0 \in \mathscr{H}_{\mathrm{univ}}$ *of the equation of motion* $Q_B\Psi + \Psi*\Psi = 0$ *whose action is* $S[\Psi_0] = \mathscr{T}_p\,\mathrm{Vol}$*, that is, whose energy density is* $-\mathscr{T}_p$ *[78].*

(ii) The kinetic operator around Ψ_0*,*

$$Q_{\Psi_0}\Phi = Q_B\Phi + \Psi_0*\Phi - (-1)^{|\Phi|}\Phi*\Psi_0,$$

has trivial cohomology in every ghost number: there is a string field A *with* $Q_{\Psi_0}A = 1$*, the identity of the star algebra [36].*

(iii) For every boundary conformal field theory BCFT_* *with the same bulk theory as* BCFT_0 *and admitting boundary condition changing operators between the two of weight zero with regular operator products, there is a solution* Ψ_* *of the equation of motion on the Dp-brane whose energy is that of* BCFT_* *and whose fluctuation spectrum is that of* BCFT_**; the tachyon lump describing a* $D(p-1)$*-brane at any compactification radius is of this form [38].*

Proof. (i) is the solution of [78] in the gauge $\mathscr{B}_0\Psi = 0$, with the energy computed there and confirmed by [69] and by [43], who checked that the equation of motion holds when contracted with the solution itself; the simpler solution of [39], $\Psi = \frac{1}{\sqrt{1+K}}\,c(1+K)Bc\,\frac{1}{\sqrt{1+K}}$ in the KBc algebra, has the same energy by a two-line computation. (ii) is [36]: for the Erler–Schnabl solution the homotopy operator is $A = B/(1+K)$, and $Q_\Psi A = 1$ is an identity of the KBc algebra. (iii) is [38, Section 4]: the solution is $\Psi_* = \Psi_{\mathrm{tv}} - \Sigma\Psi_{\mathrm{tv}}\bar{\Sigma}$ with $\Sigma, \bar{\Sigma}$ built from the boundary condition changing operators and a timelike Wilson line, and $\bar{\Sigma}\Sigma = 1$; the energy is the disc partition function of BCFT_* and the cohomology of Q_{Ψ_*} is isomorphic to that of BCFT_*. The hypothesis on the boundary condition changing operators is satisfied by the lump at every radius, by the Wilson line construction of the same paper; the numerical lumps of [60] agree. □

Verdict VI.4 (claims 12, 13, 14)**.** Proved. Each statement is a theorem of cubic open string field theory; the sense in which it is a theorem about string theory depends on cubic string field theory being a complete description of the open string, which is the framework of the conjectures and is not in question in them.

Theorem VI.5 (claim 15; Erler 2013)**.** *In the open superstring field theory of Berkovits on a non-BPS D-brane there is an analytic solution for the tachyon vacuum, built from* 2×2 *matrices in two* GL_2 *subgroups of the star algebra, whose action equals the tension of the non-BPS brane, so that the energy density of the vacuum is* $-\mathscr{T}_p$*.*

Proof. [37]. The energy is computed in closed form from the nonpolynomial action. □

Verdict VI.6 (claim 15)**.** Proved for the energy. The absence of open string states around the superstring vacuum follows from the homotopy operator of the cubic reformulation of the same solution, and the lower-dimensional branes in the superstring are not yet covered by an analytic solution; the numerical results of [12], the level truncation to level $(2,4)$ with 85% of the tension, were the state of the conjecture before [37]. Of the papers citing [37], none constructs a homotopy operator for the kinetic operator around Erler's solution or a lump solution of the Berkovits theory; the second and third conjectures remain open there, and the first is proved.

VII. THE ROLLING TACHYON AND TACHYON MATTER

A. Homogeneous decay

Sen [85] described the time-dependent decay of an unstable Dp-brane by the boundary conformal field theory with the marginal deformation $\lambda\int\cosh X^0$ of the free boson X^0 on the boundary of the world-sheet, and computed the energy-momentum tensor from the boundary state. In the bosonic

string, for $0 \le \lambda \le \frac{1}{2}$,

$$T_{00} = \tfrac{1}{2}\mathscr{T}_p\big(1+\cos 2\pi\lambda\big), \qquad T_{ij} = -\delta_{ij}\,\mathscr{T}_p f(x^0),$$
$$f(x^0) = \frac{1}{1+e^{x^0}\sin\pi\lambda} + \frac{1}{1+e^{-x^0}\sin\pi\lambda} - 1, \tag{10}$$

and in the superstring the same with x^0 replaced by $x^0/\sqrt{2}$ and $\sin\pi\lambda$ by $\sin^2\pi\lambda$.

Theorem VII.1 (claim 16; Sen 2002)**.** *The energy density* (10) *is conserved, and the pressure* $-\mathscr{T}_p f(x^0)$ *tends to zero as* $x^0 \to \pm\infty$*, as* $e^{-|x^0|}$ *in the bosonic string and* $e^{-|x^0|/\sqrt{2}}$ *in the superstring. For* $\lambda = \frac{1}{2}$ *the energy density vanishes and the solution is the tachyon vacuum at all times.*

Proof. The boundary state is the product of the Dirichlet boundary state for the spatial directions, the $c = 1$ boundary state of the $\cosh X^0$ deformation, obtained by Wick rotation of the $SU(2)$ rotation of the Neumann boundary state at the self-dual radius, and the ghost boundary state; $T_{\mu\nu}$ is read off from the coefficient of the closed string graviton. As $x^0 \to +\infty$ the first term of f is $O(e^{-x^0})$ and the second is $1 - \sin\pi\lambda\, e^{-x^0} + O(e^{-2x^0})$, so $f = O(e^{-x^0})$; the case $x^0 \to -\infty$ is symmetric. Conservation is $\partial_0 T_{00} = 0$, immediate from the form of T_{00}. This is [85, Section 3]; the computation is exact in α' and at tree level in the string coupling. □

Verdict VII.2 (claim 16)**.** Proved. The statement is a theorem of boundary conformal field theory. The open question is its physical scope, treated in Section VII C.

B. Tachyon matter and the caustic theorem

Sen [84] called the asymptotic state of (10), with constant energy density and vanishing pressure, *tachyon matter*, and [83] proposed an effective field theory for the tachyon near the minimum of its potential whose solutions are this matter:

$$S = -\int d^{p+1}x\, V(T)\sqrt{1+\eta^{\mu\nu}\partial_\mu T\partial_\nu T}, \qquad \eta = \mathrm{diag}(-1,1,\dots,1), \quad V(T)\to 0 \text{ as } T\to\infty. \tag{11}$$

The abstract of [84] states the conclusion: "in both cases the asymptotic form of the energy momentum tensor for large time is described by a pressureless gas with non-zero energy density", the two cases being the bosonic string and the superstring; and [83] states that the classical solutions of (11) near the minimum "include non-relativistic matter with arbitrary spatial distribution of energy". Both sentences are proved in [84, 85] for spatially homogeneous decay, and the second is correct as a statement about the existence of solutions at one instant. Read as a statement about the end state of the decay of a brane whose tachyon is not homogeneous, which is how the two papers and much of the literature read it, the first sentence is false, and the reason is in the equations of [83] themselves.

Near the minimum, $V \to 0$, the equation of motion of (11) forces $\eta^{\mu\nu}\partial_\mu T\partial_\nu T \to -1$ wherever the energy density stays finite [83, Section 2], and the energy-momentum tensor takes the form

$$T^{\mu\nu} = \rho\, u^\mu u^\nu, \qquad u^\mu = -\eta^{\mu\nu}\partial_\nu T, \qquad u^\mu u_\mu = -1, \tag{12}$$

of a pressureless fluid with velocity u^μ and rest density $\rho = V(T)/\sqrt{1+\partial T\cdot\partial T} \ge 0$. We write $W = \partial_t T = \sqrt{1+|\nabla T|^2}$ for the branch in which T increases with time (the other branch is obtained by $T \mapsto -T$), $v = -\nabla T/W$ for the three-velocity u^i/u^0 of the fluid, and $\varepsilon = T^{00} = \rho W^2$ for the energy density.

Lemma VII.3. *Let* T *be a* C^2 *solution of* $\eta^{\mu\nu}\partial_\mu T\partial_\nu T = -1$ *on* $[0,t_1)\times\mathbb{R}^p$ *with* $\partial_t T > 0$*, and let* $\rho \ge 0$ *be* C^1 *with* $\partial_\mu(\rho u^\mu) = 0$*. Then:*

(i) $u^\mu\partial_\mu u^\nu = 0$*: the integral curves of* u*, equivalently the curves* $\dot X = v(t,X)$*, are straight lines* $X(t;x_0) = x_0 + t\,v_0(x_0)$*,* $v_0 = v(0,\cdot)$*, along which* v *is constant;*

(ii) $\partial_t\varepsilon + \partial_i(\varepsilon v^i) = 0$*, and along the lines of (i)*

$$\varepsilon\big(t,X(t;x_0)\big)\det\big(I + t\,Dv_0(x_0)\big) = \varepsilon(0,x_0);$$

(iii) $Dv_0 = -W^{-1}PH$ *with* $H = \nabla^2 T(0,\cdot)$ *the Hessian of the initial profile and* $P = I - \nabla T\nabla T^{\mathsf T}/W^2$ *symmetric and positive definite; the eigenvalues of* $Dv_0(x_0)$ *are real, and* $Dv_0(x_0)$ *has a negative eigenvalue if and only if* $H(x_0)$ *has one.*

Proof. (i) Since $u_\mu = -\partial_\mu T$ is a gradient,

$$u^\mu\partial_\mu u_\nu = u^\mu\partial_\nu u_\mu = \tfrac{1}{2}\partial_\nu(u^\mu u_\mu) = 0.$$

In components, with $u^0 = W$ and $u^i = Wv^i$, the equations $u^\mu\partial_\mu W = 0$ and $u^\mu\partial_\mu(Wv^i) = 0$ give $(\partial_t + v\cdot\nabla)v = 0$, so v is constant along the curves $\dot X = v$, which are therefore straight. (ii) $\partial_\mu(\rho u^\mu u^0) = u^0\partial_\mu(\rho u^\mu) + \rho u^\mu\partial_\mu u^0 = 0$ is the continuity equation for $\varepsilon = \rho W^2$ with flux εv. Writing it as $(\partial_t + v\cdot\nabla)\varepsilon = -\varepsilon\,\nabla\cdot v$ and using

$$\nabla\cdot v\big(t,X(t;x_0)\big) = \tfrac{d}{dt}\ln\det\big(I + tDv_0(x_0)\big),$$

which holds because the Jacobian $D_{x_0}X = I + tDv_0$ satisfies $\frac{d}{dt}D_{x_0}X = (Dv)(D_{x_0}X)$, gives

$$\tfrac{d}{dt}\big[\varepsilon\det(I+tDv_0)\big] = 0$$

along the line. (iii) Differentiating $v^i = -\partial_i T/W$ and $W^2 = 1+|\nabla T|^2$ gives $\partial_j v^i = -W^{-1}\big(\delta_{ik} - \partial_i T\partial_k T/W^2\big)\partial_k\partial_j T$. The matrix P has eigenvalue 1 on the orthogonal complement of ∇T and $1 - |\nabla T|^2/W^2 = W^{-2} > 0$ along it. PH is similar to $P^{1/2}HP^{1/2}$, which is symmetric, so its eigenvalues are real, and by Sylvester's law of inertia $P^{1/2}HP^{1/2}$ and H have the same number of negative eigenvalues. □

Theorem VII.4 (claim 17 fails as stated)**.** *Let* $T_0 \in C^2(\mathbb{R}^p)$ *be the initial profile of a solution of the dust equations* (12) *of*

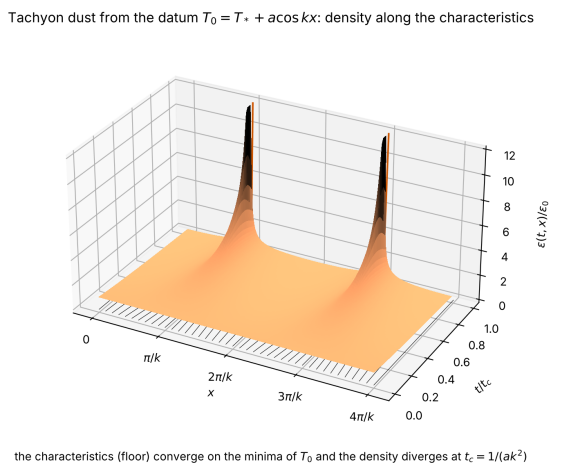


FIG. 3. The dust of Theorem VII.4 for the datum $T_0 = T_* + a\cos kx$ of Appendix B, with $a = 1/2$, $k = 1$. The characteristics, drawn on the floor, are straight lines that converge on the minima of T_0; the energy density along them is $\varepsilon_0/(1 + t\,v_0'(x_0))$ and diverges at $t_c = 1/(ak^2)$, where the vertical lines stand. The surface is cut at twelve times the initial density.

[83], on the branch $\partial_t T > 0$, with initial energy density $\varepsilon_0 > 0$. Let

$$\mu_- = \sup_{x_0 \in \mathbb{R}^p} \max\big\{ -\lambda : \lambda \text{ an eigenvalue of } Dv_0(x_0) \big\}, \qquad Dv_0 = -\frac{P\nabla^2 T_0}{\sqrt{1 + |\nabla T_0|^2}}. \tag{13}$$

(i) If T_0 is convex, $\mu_- \le 0$: along every characteristic the Jacobian $\det(I + tDv_0(x_0))$ is at least 1 for all $t \ge 0$, the energy density is non-increasing, and no caustic forms.

(ii) If T_0 is not convex, then $0 < \mu_- \le \infty$ and the solution develops a caustic at the time $t_c = 1/\mu_-$: two characteristics meet, the Jacobian $\det(I + tDv_0)$ vanishes along some characteristic, and the energy density ε along that characteristic tends to $+\infty$ as $t \uparrow t_c$. The solution of the dust equations does not extend past t_c as a single-valued field.

(iii) Every bounded non-constant T_0, in particular every periodic or compactly supported non-constant perturbation of the homogeneous decay, falls under (ii).

(iv) For $p = 1$: $v_0' = -T_0''/(1 + T_0'^2)^{3/2}$, the caustic forms if and only if $T_0'' > 0$ somewhere, $t_c = \big(\sup_x T_0''(x)/(1 + T_0'(x)^2)^{3/2}\big)^{-1}$, and $\varepsilon(t, X(t;x_0)) = \varepsilon_0(x_0)/(1 + t\,v_0'(x_0))$.

Proof. By Lemma VII.3(iii) the eigenvalues of $Dv_0(x_0)$ are real, and are all nonnegative at x_0 exactly when $\nabla^2 T_0(x_0) \succeq 0$. (i) If T_0 is convex this holds at every x_0, so $\det(I + tDv_0(x_0)) = \prod_k (1 + t\lambda_k(x_0)) \ge 1$ and is non-decreasing for $t \ge 0$, and Lemma VII.3(ii) gives $\varepsilon(t, X(t;x_0)) \le \varepsilon_0(x_0)$, non-increasing in t. (ii) If T_0 is not convex there is x_0 at which $\nabla^2 T_0$ has a negative eigenvalue, hence $Dv_0(x_0)$ has an eigenvalue $\lambda < 0$, and $\det(I + tDv_0(x_0))$ vanishes at $t = -1/\lambda$. The infimum of these first zeros over x_0 is $t_c = 1/\mu_-$ (with $t_c = 0$ if $\mu_- = \infty$, which happens only for unbounded curvature). For $t < t_c$ the solution exists along every characteristic, and along a characteristic from a point x_0 at which the first zero is attained or approached, Lemma VII.3(ii) gives $\varepsilon(t, X(t;x_0)) = \varepsilon_0(x_0)/\det(I + tDv_0(x_0)) \to +\infty$ as $t \uparrow t_c$; if the supremum is only approached, the same holds along a sequence of characteristics. At t_c the map $x_0 \mapsto X(t_c;x_0)$ is not a local diffeomorphism, and for $t > t_c$ two characteristics carrying different velocities pass through the same point, so v, and therefore ∇T, is not single valued. (iii) A convex function on $\mathbb{R}^p$ that is bounded above is constant: along any line it is a convex function of one variable bounded above, hence constant. So a bounded non-constant T_0 is not convex. (iv) is (ii) with $P = (1 + T_0'^2)^{-1}$ and $H = T_0''$. Appendix B and Figure 3 work out the time t_c and the density profile for the perturbation $T_0 = a\cos kx$ of the homogeneous decay and compares them with the numerical solutions of [41]. □

Verdict VII.5 (claim 17)**.** False as stated. Tachyon matter is the final state of homogeneous decay, by Theorem VII.1, and is not the final state of inhomogeneous decay: by Theorem VII.4 the dust of [83] reaches infinite density at a finite time t_c for every initial profile that is bounded and not constant, and the equations stop describing anything after t_c. The caustics were found numerically in the full Born–Infeld equation (11) by Felder, Kofman and Starobinsky [41], who also derived the free-streaming approximation, and analytically for several tachyon effective actions by Barnaby [4] and Felder and Kofman [40]; the theorem above is the exact statement for the dust limit of [83], with the convexity criterion and the time. What the caustic becomes in string theory is open: the effective action (11) is derived for slowly varying fields and is not valid where second derivatives diverge, and the two candidates for the physics there, the formation of closed strings [63] and the appearance of higher-derivative corrections that smooth the fold, are not decided. Sen's review [88] states the result for the homogeneous solution, which is what is proved; the general sentence of [84] is what we record as false.

C. Closed string emission and open–closed duality

The decaying brane couples to closed strings, and the question whether the classical open string description of Theorem VII.1 survives their emission was raised in [85] and answered in part by Lambert, Liu and Maldacena [63]: the energy radiated into closed strings by a decaying Dp-brane is infinite at leading order for $p \le 2$ and finite for $p > 2$; for $p = 0$ the divergence, cut off at energies of order $1/g_s$, carries away an energy of the order of the mass of the brane, so that all the energy of the brane goes into closed strings. Sen [87] turned this into the conjecture of *open–closed duality at tree level*: the tree-level open string theory on the decaying brane, with its tachyon matter, is a dual description of the closed string state produced by the decay, so that the two are

not to be added.

Verdict VII.6 (claim 18)**.** Confirmed. The conjecture is supported by the computation of [63], in which the closed string state produced is determined by the open string boundary state and its energy, for $p = 0$, is the energy of the brane; by the matching of the energy-momentum tensor of the emitted closed strings with that of tachyon matter in the regime where both can be computed [87]; and by the two-dimensional string theory, where the decay of the D0-brane is the motion of a single fermion of the matrix model and the open and closed descriptions are provably the same [88]. No proof exists in the critical string, and the missing step is a control of the backreaction of the emitted closed strings on the open string dynamics beyond leading order in g_s. The conjecture is stated for homogeneous decay; after Theorem VII.4 the inhomogeneous case has no classical open string side to be dual to beyond t_c.

VIII. NON-BPS BRANES AND THE DESCENT RELATIONS

The tachyon conjectures of Section VI have a static counterpart: the solitons of the tachyon field are themselves D-branes. Sen [116] identified the vortex on a Dp–$\overline{\text{D}}p$ pair of type II with the BPS D$(p-2)$-brane, [119] identified the kink on a D1–$\overline{\text{D}}$1 pair of type I with a stable non-BPS D-particle transforming as a spinor of $SO(32)$, the state required by the duality with the heterotic string, and [120] organised these into the *descent relations*: the non-BPS Dp-brane of type II is the orbifold of the Dp–$\overline{\text{D}}p$ pair by $(-1)^{F_L}$, and the BPS D$(p-1)$-brane is the kink on the non-BPS Dp-brane.

Theorem VIII.1 (claims 19, 20)**.** *(i) At the critical radius $R = \sqrt{2}$ (in units $\alpha' = 1$) of a circle transverse to the brane pair, the tachyon kink on a Dp–$\overline{\text{D}}p$ pair of type II wrapped on the circle is an exactly marginal boundary deformation, and the boundary conformal field theory at the end of the deformation is that of a non-BPS D$(p-1)$-brane; the kink on the non-BPS Dp-brane at $R = 1$ is, in the same way, the BPS D$(p-1)$-brane.*

(ii) The conserved charges of D-branes and of their tachyon solitons are classified by the K-theory of spacetime: $K(X)$ for type IIB, $K^{-1}(X)$ for type IIA, $KO(X)$ for type I; the kink on the type I D1–$\overline{\text{D}}$1 pair carries the class generating $KO^{-1}(\text{pt}) = \mathbb{Z}_2$ and is the $SO(32)$ spinor, stable and non-BPS.

Proof. (i) is the construction of [119, 120]: the tachyon of the wrapped pair at the critical radius has a vertex operator $\cos(X/\sqrt{2}) \otimes \sigma_1$ which is exactly marginal, the deformation by it can be followed to its end by fermionising the boson X, and the boundary state at the end is the boundary state of the lower brane with Dirichlet condition on X. (ii) is Witten's theorem [126], which starts from Sen's identification of the brane–antibrane system with a pair of bundles (E,F) and the tachyon with a map between them and identifies the charge with the class of (E,F) in K-theory; the type I computation $KO^{-1}(\text{pt}) = \mathbb{Z}_2$ gives the $\mathbb{Z}_2$ charge of the spinor, which prevents its decay. □

Verdict VIII.2 (claims 19, 20)**.** Proved. Both statements are theorems of boundary conformal field theory and of K-theory respectively; the existence of the spinor as a stable non-BPS state was confirmed by its mass at weak coupling, which matches the heterotic spinor's at strong coupling up to the expected corrections, and by the identification of the stable non-BPS states of other orbifolds and orientifolds by the same method [121].

Proposition VIII.3 (claim 21; Sen 2003)**.** *In the effective action (11) on a non-BPS Dp-brane, the kink solution $T(x) = \lim_{a\to\infty} ax$ (a step of infinite slope) is singular as a field and regular as a source: its energy-momentum tensor is $\mathscr{T}_{p-1}$ times a delta function at $x = 0$, and the effective action of its collective coordinates and world-volume gauge field is exactly the Dirac–Born–Infeld action of a BPS D$(p-1)$-brane, to all orders in the fields and for every potential V with $\int V\,dT$ finite. The same holds for the vortex on the brane–antibrane pair.*

Proof. [86], Sections 3 and 4: for the ansatz $T = f(a(x - \xi(y)))$ with f odd and increasing, in the limit the action evaluated on the ansatz is independent of the profile f and of the scale a, by the change of variables $T = f(\cdot)$, and equals $-\mathscr{T}_{p-1}\int\sqrt{-\det(g_{\alpha\beta}+F_{\alpha\beta})}$ with the induced metric of the embedding ξ; the equations of motion of the ansatz follow from the full equations because the action is stationary in f and a. □

Verdict VIII.4 (claim 21)**.** Proved, within the effective action. The statement is about (11) and is exact there; it says nothing about the string theory corrections to (11), which are not controlled on the kink, where second derivatives are large. The agreement of the tension and of the form of the world-volume action with those of the D$(p-1)$-brane is the content of the claim and holds.

IX. THE SEN LIMIT OF F-THEORY

Sen [114] showed that F-theory on an elliptic K3 surface, at the point of its moduli space where the Weierstrass model $y^2 = x^3 + f(z)x + g(z)$ has constant modulus τ and four D_4 singular fibres, is the type IIB orientifold on $T^2/\mathbb{Z}_2$ with four orientifold seven-planes and four D7-branes on each, with gauge group $SO(8)^4$ and constant coupling; and that moving away from this point, in a one-parameter family $f = -3h^2 + \varepsilon\eta$, $g = -2h^3 + \varepsilon h\eta - \frac{1}{12}\varepsilon^2\chi$ with $\varepsilon \to 0$, the coupling becomes weak away from the loci $h = 0$, which are the orientifold planes, while the discriminant $\Delta = -9\varepsilon^2 h^2(\eta^2 - h\chi) + O(\varepsilon^3)$ places the D7-branes on $\eta^2 = h\chi$. In [115] the same expansion was applied to F-theory on elliptic Calabi–Yau threefolds and fourfolds over a base B, with h, η, χ sections of $K_B^{-2}, K_B^{-4}, K_B^{-6}$: the statement is that F-theory on such a Calabi–Yau has a weak coupling limit which is type IIB on the double cover of

B branched along $h=0$, with the orientifold projection, O7-planes on $h=0$ and D7-branes on $\eta^2=h\chi$.

Theorem IX.1 (claim 22; Sen 1996)**.** *F-theory on the elliptic K3 with Weierstrass model $y^2=x^3+fx+g$, $f=-3h^2+\varepsilon\eta$, $g=-2h^3+\varepsilon h\eta-\varepsilon^2\chi/12$, h of degree four in z, has, as $\varepsilon\to 0$, the modulus $\tau(z)\to i\infty$ away from $h=0$, with $j(\tau)\sim h^4/(\varepsilon^2(\eta^2-h\chi))$, and the monodromy around $h=0$ is $-1\in SL(2,\mathbb{Z})$ at $\varepsilon=0$: the limit is the type IIB orientifold $T^2/\mathbb{Z}_2$ with the $\mathbb{Z}_2$ acting as $(-1)^{F_L}\Omega$ composed with the reflection of T^2, four O7-planes at the zeros of h and sixteen D7-branes at the roots of $\eta^2=h\chi$, which collide in groups of four with the O7-planes at $\eta=\chi=0$ to give the $SO(8)^4$ point.*

Proof. [114], Sections 2 to 4. The monodromy computation uses that at $\varepsilon=0$ the fibration $y^2=(x-h)^2(x+2h)$ has I_0^*-type degeneration along $h=0$ whose monodromy is -1, and that for $\varepsilon\neq 0$ this splits into the -1 of the orientifold plane (two I_1 fibres with monodromies whose product is the Dehn twist squared composed with -1) and the I_1 fibres of the D7-branes. The identification of the charges, four D7 per O7, is forced by the degree of Δ (24) and the charge -4 of each O7. □

Verdict IX.2 (claim 22)**.** Proved. The statement is exact at the $SO(8)^4$ point, where the modulus is constant, and the limit $\varepsilon\to 0$ is the stable degeneration of the K3 into two rational elliptic surfaces [17], with the orientifold as the double curve.

Proposition IX.3 (claim 23)**.** *Let B be a smooth base of an elliptic Calabi–Yau threefold or fourfold with the Sen family of Weierstrass models. Then:*

(i) the D7-brane locus $\eta^2=h\chi$ is a divisor of B with a double point locus along $\eta=h=0$ (and, for fourfolds, a pinch point locus along $\eta=h=\chi=0$): it is the Whitney umbrella, not a smooth divisor, and the O7-plane $h=0$ is tangent to it along the double curve [15];

(ii) the limit is a stable degeneration of the Calabi–Yau to the union of two $\mathbb{P}^1$-fibred components over B, glued along the double cover X of B branched along $h=0$, with the D7-brane data encoded in a spectral cover [17];

(iii) for bases B with non-Higgsable clusters, that is, divisors of self-intersection ≤ -3 in a surface base or their analogues in threefold bases, the generic Weierstrass model has non-abelian gauge symmetry on them for every value of the complex structure, Δ vanishes to order at least two on them, and the family $f=-3h^2+\varepsilon\eta$ does not reach a weak coupling limit in which these divisors carry D7-branes of a perturbative orientifold: the Sen limit does not exist as a perturbative type IIB orientifold on such bases [66, 67]. For surface bases these include the Hirzebruch surfaces $\mathbb{F}_n$, $n\geq 3$, and most toric bases; for threefold bases almost all.

Proof. (i) is a computation: near a point of $\eta=h=0$ with $\chi\neq 0$, the equation $\eta^2=h\chi$ is $\eta^2=h$ after rescaling, the Whitney umbrella $\{u^2=vw\}$ in the fourfold case; its singular locus and the tangency are read off from the equations [15, Section 2]. (ii) is [17, Sections 3 and 4], by the semistable reduction of the Weierstrass family. (iii) is the classification of [66] for surface bases and [67] for threefold bases: on a curve of self-intersection $-n\leq -3$ the sections f,g vanish to orders ($\lceil 2-4/n\rceil$-type) forced by the adjunction formula, the fibre is of Kodaira type IV, I_0^*, IV^*, III^* or II^* for every complex structure, and the coupling on such a curve is fixed at a finite value, so no family of the Sen type sends $\tau\to i\infty$ there. □

Verdict IX.4 (claim 23)**.** Incomplete. The statement of [115] that the orientifold limit of F-theory on a Calabi–Yau threefold or fourfold places the D7-branes on a smooth divisor $\eta^2=h\chi$ of the base is false as a statement about the divisor, which is singular along $\eta=h=0$ [15], and the sentence that the limit exists for any elliptic Calabi–Yau is false on bases with non-Higgsable clusters [66, 67]. The construction of [115] is correct where it applies: on bases without non-Higgsable clusters the degeneration exists [17], and the singular D7-brane is a feature of the limit, not an error in it, once the tangency with the O7-plane is understood; the tadpole and the spectrum computed in [115] for $\mathbb{P}^2$ and $\mathbb{F}_0$, $\mathbb{F}_1$, $\mathbb{F}_2$ bases are correct. We mark the scope, not the computation.

X. BLACK HOLE ENTROPY: THE ENTROPY FUNCTION AND SMALL BLACK HOLES

A. The entropy function

For an extremal black hole in a theory of gravity with arbitrary higher-derivative terms, whose near-horizon geometry has the symmetry $SO(2,1)\times SO(D-1)$ of $AdS_2\times S^{D-2}$ with electric fields e_i and magnetic charges p_a, Sen [90] defined the entropy function

$$\begin{gathered}\mathscr{E}(\vec{u},\vec{v},\vec{e},\vec{q},\vec{p})=2\pi\big(e_iq_i-f(\vec{u},\vec{v},\vec{e},\vec{p})\big),\\ f=\int_{S^{D-2}}\sqrt{-\det g}\,\mathscr{L},\end{gathered}\tag{14}$$

where f is the Lagrangian density integrated over the sphere on the near-horizon ansatz with radii v_1,v_2 of AdS_2 and S^{D-2} and constant scalars u_s.

Theorem X.1 (claims 24, 25)**.** *(i) The equations of motion of the near-horizon geometry are the conditions*

$$\partial\mathscr{E}/\partial u_s=\partial\mathscr{E}/\partial v_{1,2}=0,\qquad \partial\mathscr{E}/\partial e_i=0,$$

the last being the definition of the charges q_i, and the Wald entropy of the black hole is the value of $\mathscr{E}$ at the extremum [90].

(ii) For rotating extremal black holes, whose near-horizon geometry has the symmetry $SO(2,1)\times U(1)$ and depends on the polar angle through functions $\Omega(\theta),\psi(\theta),\ldots$, the same holds with $\mathscr{E}$ a functional of these functions and with an additional term $2\pi\alpha J$ for the angular momentum, the extremisation being over the functions as well as the constants [1].

Proof. (i) The Wald entropy is $S=-8\pi\int_H \frac{\partial\mathscr{L}}{\partial R_{\mu\nu\rho\sigma}}\varepsilon_{\mu\nu}\varepsilon_{\rho\sigma}$; on the $AdS_2\times S^{D-2}$ ansatz the $SO(2,1)$ symmetry makes every covariant tensor built from the fields a combination of the metric and the volume forms with constant coefficients, and the Wald integrand reduces to $v_1\partial f/\partial v_1$ evaluated at the extremum, which by the Euler equations for v_1 and e_i and the homogeneity of f equals $2\pi(e_iq_i-f)$ [90, Section 2]. The equations of motion reduce to the extremisation because the ansatz is the most general one with the symmetry, so that the variation of the action over field configurations with the symmetry is the variation over the ansatz (the principle of symmetric criticality, valid here because the symmetry group is compact times $SO(2,1)$ acting with the stated isotropy). (ii) is [1, Section 2], with the same argument applied to the $SO(2,1)\times U(1)$ ansatz; the extra term arises because the $U(1)$ gauge field of the Kaluza–Klein reduction along the angle carries the angular momentum as its charge. □

Verdict X.2 (claims 24, 25)**.** Proved. The theorem is a derivation within classical higher-derivative gravity with the stated symmetry; the attractor mechanism, that the entropy is independent of the asymptotic moduli, is a corollary whenever the extremum is isolated in the scalars. The results agree with the earlier supergravity computations of Cardoso, de Wit and Mohaupt where both apply [90], and the formalism has been used in every later comparison of macroscopic and microscopic entropies.

B. Small black holes

A half-BPS state of the heterotic string on T^6 with winding w and momentum n on a circle, with $nw>0$, has $d_h(nw)\sim e^{4\pi\sqrt{nw}}$ states, while the two-derivative supergravity solution with these charges has a singular horizon of zero area. Sen [89] argued that the higher-derivative terms stretch the horizon to string scale and that the entropy function with the Gauss–Bonnet term, $\mathscr{L}\supset\frac{1}{16\pi G}\phi(S)\big(R^2_{\mu\nu\rho\sigma}-4R^2_{\mu\nu}+R^2\big)$ with the one-loop coupling $\phi(S)$, gives

$$S_{\rm BH}=4\pi\sqrt{nw} \tag{15}$$

in four dimensions, in agreement with the statistical entropy to leading order; [91] extended the computation to the heterotic string in D dimensions.

Proposition X.3 (claim 26)**.** *With the Gauss–Bonnet term as the only higher-derivative correction, the entropy function gives a finite entropy for the small black hole in every dimension $D\geq 4$, proportional to $\sqrt{nw}$. The coefficient equals the statistical one, 4π, for $D=4$ and $D=5$, and differs from it for every $D\geq 6$ [91, Section 4]. In four dimensions the agreement extends to all orders in $1/\sqrt{nw}$ when the full supersymmetric F-term corrections are used in place of the Gauss–Bonnet term [22, 26].*

Proof. The near-horizon solution of [91] has v_1,v_2 and the dilaton fixed by the extremisation with the Gauss–Bonnet coupling, and the entropy at the extremum is $c_D\sqrt{nw}$ with c_D computed there: $c_4=c_5=4\pi$ and $c_D\neq 4\pi$ for $D\geq 6$ (the paper tabulates the ratio). For $D=4$, [22] evaluated the entropy with the full F-term, and [26] matched the result to the asymptotic expansion of d_h to all orders in $1/\sqrt{nw}$ up to the non-perturbative terms; the Gauss–Bonnet term alone reproduces the leading term because the four-dimensional F-term reduces to it on the near-horizon geometry at this order. □

Verdict X.4 (claim 26)**.** Incomplete. The sentence of [89] that the Gauss–Bonnet term reproduces the entropy of small black holes, taken as a statement about the heterotic string in every dimension, holds in four and five dimensions only; Sen's own computation [91] shows the mismatch for $D\geq 6$ and attributes it to the Gauss–Bonnet term not being the full supersymmetric completion in those dimensions. The four-dimensional claim is proved to all orders in $1/\sqrt{nw}$ [22, 26]. The remaining issue, that the small black hole is not a solution of the two-derivative theory and the higher-derivative expansion is not under control at string scale curvature, was raised in [26] and is unresolved; the agreement of (15) is a fact about the Gauss–Bonnet action, and its interpretation as the entropy of a stretched horizon is the conjecture.

XI. THE QUANTUM ENTROPY FUNCTION AND THE LOGARITHMIC CORRECTIONS

A. The quantum entropy function

Sen [95] proposed that the exact degeneracy of the horizon states of an extremal black hole with charges $\vec{q}$ is the finite part of the Euclidean string theory path integral on the near-horizon AdS_2,

$$d_{\rm hor}(\vec{q})=\Big\langle \exp\Big[-iq_i\oint A^{(i)}_\theta\,d\theta\Big]\Big\rangle^{\rm finite}_{AdS_2}, \tag{16}$$

with a Wilson line insertion at the boundary carrying the charges, the boundary conditions fixing the charges rather than the chemical potentials, and the "finite part" defined by subtracting the terms proportional to the length of the boundary; and that this quantity equals the microscopic index B_6 for quarter-BPS black holes, after the hair modes are removed (Section V C).

Verdict XI.1 (claim 27)**.** Confirmed. The proposal passes every test that can be computed: the classical limit of (16) is the exponential of the Wald entropy; the one-loop correction is the logarithmic term, which matches the microscopic index exactly in $\mathscr{N}=4$ and $\mathscr{N}=8$ (Theorem XI.2); and the localisation of the path integral in $\mathscr{N}=2$ supergravity, proposed in [5] and carried out in [29, 30], reproduces the exact Rademacher expansion of the index of $\mathscr{N}=8$ black holes and the leading Bessel function of the $\mathscr{N}=4$ one [51, 52, 68]. The last of these results is weaker than its statement in the literature, and Sen's own later assessment [110] says so: the claim that supersymmetric localisation of the AdS_2 path integral reproduces the Rademacher expansion from first principles rests on three steps that are not established. The Killing

spinors of the near-horizon geometry grow at the boundary of AdS_2 and do not act on the space of normalisable fluctuations over which the path integral is defined, so the localisation argument has no proved domain; the measure on the localisation manifold used in [29, 68] is postulated and not derived, and a different measure changes every subleading term; and the one-loop determinants over the transverse directions depend on an ultraviolet cut-off that the computation sets, without justification, at the four-dimensional Planck scale. Each of these is listed in [110], Sections 3 to 5 and 8, as open. There is no proof: the path integral is not defined beyond the semiclassical expansion, the localisation argument assumes an off-shell supersymmetric completion of the higher-derivative terms, and the identification of the subtracted terms with the hair is made case by case. The missing step is a definition of the right-hand side of (16) that does not rely on a saddle point expansion; the conjecture is, in this sense, a proposal for what that definition should give.

B. The logarithmic corrections

The one-loop contribution to (16) from the massless fields on $AdS_2 \times S^2$ gives a term $c \ln A_H$ in the entropy, with c determined by the spectrum and the two-derivative action alone. Banerjee, Gupta, Mandal and Sen [6] computed it for quarter-BPS black holes in $\mathcal{N}=4$ and eighth-BPS black holes in $\mathcal{N}=8$ string theory, and Sen [98] for half-BPS black holes in $\mathcal{N}=2$ theories with n_V vector and n_H hypermultiplets.

Theorem XI.2 (claim 28)**.** *The logarithmic correction to the entropy of a quarter-BPS black hole in any $\mathcal{N}=4$ string compactification is zero, and that of an eighth-BPS black hole in $\mathcal{N}=8$ string theory on T^6 is $-4\ln A_H$. Both agree with the microscopic indices: the asymptotic expansion of the Fourier coefficients of $1/\Phi_{10}$ and its CHL analogues has no logarithmic term, and that of the $\mathcal{N}=8$ index, a Fourier coefficient of $\vartheta_1(\tau,z)^2/\eta(\tau)^6$, has the term $-4\ln\Delta$ with $\Delta \propto A_H^2$, that is, $-2\ln\Delta = -4\ln A_H$ [6].*

Proof. The macroscopic computation is the heat kernel of the quadratic fluctuation operator of all massless fields on $AdS_2 \times S^2$ with the near-horizon flux, including the graviton, the gravitini and the vector fields, with the zero modes of the asymptotic symmetries treated separately; the coefficient is the s^0 term of the heat kernel together with the zero-mode counting, $-(c_{\mathrm{local}} + c_{\mathrm{zm}})\ln a$ with a the radius. In $\mathcal{N}=4$ the contributions cancel between the gravity multiplet and the vector multiplets [6, Section 7]; in $\mathcal{N}=8$ the net result is $-8\ln a = -4\ln A_H$ [6, Section 8]. The microscopic expansions are in [6, Section 2], from the saddle point expansion of the inverse Siegel form [25] and of the Jacobi form. The computation was rederived from the on-shell action of the Euclidean solution and from the AdS_2 Schwarzian in [51, 52], with the same coefficients. □

Verdict XI.3 (claim 28)**.** Proved, as a match between two computations each of which is complete: the macroscopic coefficient is a theorem of one-loop quantum field theory on the near-horizon geometry and the microscopic one of the asymptotics of modular forms. The rotating black holes in four and five dimensions [99] and the extremal Kerr and Reissner–Nordström black holes without supersymmetry [98, 100], for which no microscopic count exists, are predictions and remain confirmed in the sense of claim 29.

Proposition XI.4 (claim 29; Sen 2012)**.** *For a half-BPS black hole in an $\mathcal{N}=2$ supergravity with n_V vector and n_H hypermultiplets, the logarithmic correction is $\frac{1}{12}(23 + n_H - n_V)\ln A_H$, equal to $\left(2 - \frac{\chi}{24}\right)\ln A_H$ with $\chi = 2(n_V - n_H + 1)$ the Euler number of the Calabi–Yau threefold in a type II compactification.*

Proof. [98], by the same heat kernel method, the only difference from $\mathcal{N}=4$ being the absence of the cancellation; the result was rederived by [56] from the heat kernel of $\mathcal{N}=2$ supergravity on the same background. □

Verdict XI.5 (claim 29)**.** Confirmed. No microscopic count of $\mathcal{N}=2$ black hole states exists with which to compare; the comparison made in [98] is with the measure in the OSV formula, and agrees with the measure proposed by Denef and Moore if their result is extrapolated to weak topological string coupling. The missing step is a microscopic index for $\mathcal{N}=2$ black holes with an asymptotic expansion. The six papers that cite [98] and test a logarithmic correction do so in anti-de Sitter space or for the $\mathcal{N}=8$ theory; none supplies the microscopic count in $\mathcal{N}=2$ that the claim needs.

XII. STRING FIELD THEORY, D-INSTANTONS, SOFT THEOREMS AND MATRIX THEORY

A. String field theory

Two claims of Sen about string field theory itself are in the catalogue. The first, with Zwiebach [124], is that closed string field theory is background independent under infinitesimal deformations of the conformal field theory: the string field theories formulated around two nearby conformal backgrounds are related by a field redefinition. The second is the construction of superstring field theory [101–103] and the proof of its unitarity [104]: a gauge-invariant one-particle-irreducible effective action for heterotic and type II strings, free of the divergences of the picture-changing operators of earlier formulations through the use of a second, free string field of picture number $-3/2$; a Batalin–Vilkovisky master action; and the cutting rules, hence unitarity, of the resulting amplitudes.

Verdict XII.1 (claims 30, 31)**.** Proved. Claim 30 is proved in [124] for marginal deformations to first order (and to second order in the companion paper), by constructing the field redefinition from the connection on the space of conformal field theories; its extension to finite deformations is a theorem only within the formal perturbation theory of the string field and is not a claim of the paper. Claim 31 is a construction and is proved in the papers that make it: the 1PI action is gauge invariant by the main identity of the vertices, the master action satisfies the master equation, and the Cutkosky rules are

derived in [104] with the $i\varepsilon$ prescription of [71]. The reader who wants the construction in one place has the review [123]. None of this depends on a conjecture; the open questions of superstring field theory are not about these statements.

B. D-instantons

The contribution of D-instantons to string amplitudes has an overall normalisation that the world-sheet theory does not fix, because the open string path integral on the instanton has zero modes, bosonic and fermionic, whose measure is ambiguous. Sen [106, 108, 109] showed that open string field theory fixes it: the zero modes are treated by changing variables from the string field to the collective coordinates with a Jacobian computed in the field theory, the gauge zero mode of the $U(1)$ on the instanton is handled by a gauge fixing that the field theory provides, and the resulting normalisation is finite and unambiguous.

Verdict XII.2 (claim 32)**.** Confirmed. In two-dimensional string theory the normalisation computed this way [106, 107] agrees with the matrix model to the order compared, including the two-instanton contribution [11].; in type IIB string theory in ten dimensions the normalisation of the D-instanton correction to the four-graviton amplitude [108] agrees with the coefficient predicted by $SL(2,\mathbb{Z})$ invariance, the Green–Gutperle Eisenstein series. The claim that string field theory fixes the normalisation in every background is confirmed by these cases and not proved in general: the construction requires the string field theory of the instanton to be well defined, which it is in these examples, and a general statement needs a general treatment of the zero modes of the Ramond sector, which no paper has given.

C. Soft theorems

Laddha and Sen [61] proved the classical soft graviton theorem in dimensions $D > 4$, the statement that the low-frequency gravitational radiation from a scattering process is determined by the momenta and angular momenta of the incoming and outgoing objects, including the sub-subleading term, directly from the classical equations [105]; and Sahoo and Sen [77] found that in $D = 4$ the soft expansion acquires logarithmic terms, $\ln\omega$ at the subleading order, whose coefficients are again universal, and conjectured the form of the corresponding quantum terms.

Verdict XII.3 (claims 33, 34)**.** Claim 33 is proved: the classical theorem in $D > 4$ is derived in [61], and the classical logarithmic terms in $D = 4$ in [77], from the classical equations of motion with the long-range forces included. Claim 34 is confirmed: the quantum logarithmic terms were computed at one loop in [77] for scalar matter coupled to gravity and to electromagnetism and found to have the conjectured form, with the classical term and a quantum correction that depends only on the asymptotic momenta; the universality of the one-loop $\ln\omega$ terms was extended to spinning particles with the non-minimal couplings allowed by gauge and coordinate invariance by Krishna and Sahoo [58], and the classical soft factors have been pushed to all orders in the electromagnetic case [55]. The remaining conjecture is that the quantum $\ln\omega$ terms are one-loop exact, and the missing step is a proof that the two-loop and higher infrared divergences exponentiate in the way the conjecture requires; [58] states this as open, and no later paper proves it.

D. Matrix theory

Sen [117] argued, independently of Seiberg [82], that the matrix theory description of M-theory in the infinite momentum frame follows from the compactification of M-theory on a light-like circle, taken as a limit of a small space-like circle, and extended it to tori T^n: the theory of N D0-branes on T^n in a limit in which the torus and the string scale shrink. For $n \leq 3$ the limit is a maximally supersymmetric Yang–Mills theory in $n+1$ dimensions; for $n = 4, 5$ it is the $(2,0)$ theory on T^5 and the little string theory on T^5 respectively; and for $n \geq 6$ the limit is, as Sen states, "a theory without an independent definition", the degrees of freedom of the D0-branes not decoupling from gravity. The two statements H1 and H2 of the catalogue are these, and they are recorded without a verdict: the first is the Seiberg–Sen argument, which is a derivation within the assumptions of the infinite momentum frame and has no independent test for $n \geq 4$; the second is a statement about a theory that is not defined, and cannot be tested in any regime. We mention them because the paper is cited as a derivation of matrix theory and the reader should know that its content for $n \geq 6$ is a limit of a non-decoupled system, as its author says.

XIII. CONCLUSION

The premise with which this paper began was that the conjectures of Ashoke Sen are for the most part irrelevant, unproved or incomplete. The examination does not bear it out, and it would be a failure of the method to pretend otherwise. Of thirty-six claims, twenty-two are theorems, by the work of Hitchin, Kottke, Rochon, Schnabl, Ellwood, Erler, Maccaferri, Witten, Bringmann, Murthy, Rosselló and others and in several cases by Sen's own later papers; ten have passed every test that exists and lack a proof because the framework in which they are made, the path integral on AdS_2, the full open string theory, the complete soft expansion, is itself not under control; two were extended beyond the regime in which they were derived and were corrected in the literature; and two are false. The two false claims are not small. The positivity conjecture was the organising principle of a decade of work on the dyon spectrum, and its failure in the toroidal orbifolds, with the hair in every frame that causes it, changes what the conjecture is about: not Fourier coefficients of Siegel forms, but the index of the horizon after the hair is removed, which is the statement of [14] and is the one that should be tested in any new model. The identification of tachyon matter with the end of brane decay was the physical picture that the rolling

tachyon papers offered, and Theorem VII.4 shows that the picture is the picture of a set of measure zero in the space of initial conditions; what happens to an inhomogeneous decaying brane after the caustic is one of the open questions that this paper leaves.

The other open questions are, in the order of the catalogue: the L^2 cohomology of the monopole moduli spaces for $k \geq 3$ beyond the compactly supported part (claim 3); the positivity of the horizon index in $\mathcal{N}=4$ models with hair in every frame, which is proved for the two toroidal orbifolds and conjectured for none other because no other is known; the existence of a non-perturbative definition of the quantum entropy function (claim 27); a microscopic index for $\mathcal{N}=2$ black holes (claim 29); a general treatment of the Ramond zero modes in the normalisation of D-instantons (claim 32); and the exponentiation of the infrared divergences behind the quantum soft logarithms (claim 34). To these we add the two questions raised by the failures: the model-independent statement of the positivity conjecture, and the fate of the caustic.

A reader who wants to repeat the computations of this paper will find them in the archive [7], together with the Lean formalisation of the finite statements: the Fourier coefficients of Tables IV and V, the wall-crossing identities of Theorem IV.2 and the twenty-one negative indices of Theorem V.4, each computed inside the Lean kernel from the definitions of the theta functions. The statements that cannot be formalised, because they are statements about string theory and not about numbers, are the verdicts; for those the paper is the argument.

Appendix A: The computation of the Fourier coefficients

The tables of this paper are Fourier coefficients of the inverse of the product (3), computed in exact rational arithmetic. We record the steps so that the numbers can be reproduced without the program, which is in the archive [7].

The genera. The twisted elliptic genera are expanded in $q = e^{2\pi i\tau}$ and $y = e^{2\pi iz}$ from the theta functions $\theta_i(\tau,z)$ and $\eta(\tau)$ as q-series with coefficients in $\mathbb{Z}[y,y^{-1}]$, and the coefficients $c_b^{(r,s)}(4n-j^2)$ are read off from $\sum_{n,j} c_b^{(r,s)}(4n-j^2)\,q^n y^j$. For K3, $F^{(0,0)} = 8A$ with $A = \sum_{i=2}^4 \theta_i(\tau,z)^2/\theta_i(\tau,0)^2$; for the $\mathbb{Z}_2$ CHL model the genera are $F^{(0,0)} = 4A$, $F^{(0,1)} = \frac{4}{3}A - \frac{2}{3}BE_2(\tau)$, $F^{(1,0)} = \frac{4}{3}A + \frac{1}{3}BE_2(\tau/2)$, $F^{(1,1)} = \frac{4}{3}A + \frac{1}{3}BE_2((\tau+1)/2)$, with $B = \theta_1(\tau,z)^2/\eta(\tau)^6$ and $E_2(\tau) = 2E(2\tau) - E(\tau)$, where E is the Eisenstein series of weight two normalised to $1 - 24\sum_{n\geq 1}\sigma_1(n)q^n$; for the $\mathbb{Z}_2$ toroidal orbifold they are (9). The arguments $\tau/2$ and $(\tau+1)/2$ are handled by expanding in $q^{1/2}$ with the sign $(-1)^n$ on the odd powers, so every series is a Laurent series in $x = q^{1/8}$, which is the smallest fractional power that occurs.

The product. Each factor $(1 - e^{2\pi i(k'\sigma + l\rho + jv)})^{-c}$ of $1/\widetilde{\Phi}$ is expanded as a binomial series in the monomial $e^{2\pi i(k'\sigma+l\rho+jv)}$ to the order needed, and the factors are multiplied as polynomials in the three variables $e^{2\pi i\rho}$, $e^{2\pi i\sigma/N}$, $e^{2\pi iv}$ truncated at the exponents $Q^2/2 \leq L$, $NP^2/2 \leq K$ wanted in the tables. The exponent $c = \sum_s e^{2\pi isl/N} c_b^{(r,s)}(4k'l - j^2)$ is an integer for $N \leq 2$, where the root of unity is ± 1; for $N = 3,5,7$ it would require cyclotomic arithmetic, and we have not computed those models. The one factor with $k' = l = 0$, $(1 - e^{\mp 2\pi iv})^{-2}$ after the prefactor is taken into account, is the only one whose expansion depends on the chamber: in the chamber $\operatorname{Im} v < 0$ it is expanded in powers of $e^{-2\pi iv}$, in the chamber $\operatorname{Im} v > 0$ in powers of $e^{2\pi iv}$. The two expansions give the two columns of Table IV at $\pm Q\cdot P$, and their difference is the wall-crossing jump of Theorem IV.2, which the program checks for every $Q^2, P^2 \in \{2,4,6\}$ and $|Q\cdot P| \leq 4$.

The index. The coefficient of $e^{2\pi i(\rho Q^2/2 + \sigma P^2/2 + vQ\cdot P)}$ in $1/\widetilde{\Phi}$, multiplied by $(-1)^{Q\cdot P+1}$, is the index $d(Q,P)$ of (2). For the orbifolds the prefactor of the product is $e^{2\pi i(\rho+\sigma/N+v)}$ for the CHL models and $e^{2\pi i(\rho+v)}$ for the toroidal orbifolds, as in [13], and the integral over σ runs over a period of length N (Remark V.9).

The tables. Table IV is the heterotic index for $Q^2, P^2 \in \{2,4,6\}$ and $-2 \leq Q\cdot P \leq 4$; it reproduces Table 1 of [97] entry by entry. Table V is the index of the $\mathbb{Z}_2$ toroidal orbifold for $Q^2 \leq 12$, $P^2 \in \{0,2,4\}$ and $Q\cdot P \in \{0,1,2\}$; it agrees with Tables 17 to 19 of [13] where they overlap and extends them from $Q^2 = 8$ to $Q^2 = 12$. Table III lists the charges in the cone (8) with $Q^2 \leq 12$, $P^2 \leq 8$, $Q\cdot P \leq 4$ whose index is negative. The same scan over the cone of the heterotic string and of the $\mathbb{Z}_2$ CHL model, up to $Q^2 \leq 12$ (heterotic) or $NQ^2/2 \leq 6$ (CHL), $P^2 \leq 6$, $Q\cdot P \leq 4$, finds no non-positive index, and the CHL values reproduce Table 2 of [97].

Appendix B: The caustic for a periodic perturbation

This appendix works out Theorem VII.4 for the simplest inhomogeneous initial datum in one space dimension, a periodic perturbation of the homogeneous decay,

$$T_0(x) = T_* + a\cos kx, \qquad a > 0,\ k > 0, \tag{B1}$$

on the branch $\partial_t T > 0$, and records the quantities that a numerical solution of the full equation (11) can be compared with.

Velocity and its derivative. With $T_0' = -ak\sin kx$ and $W_0 = \sqrt{1 + a^2k^2\sin^2 kx}$,

$$v_0(x) = \frac{ak\sin kx}{\sqrt{1+a^2k^2\sin^2 kx}}, \qquad v_0'(x) = \frac{ak^2\cos kx}{(1+a^2k^2\sin^2 kx)^{3/2}}. \tag{B2}$$

The dust moves away from the maxima of T_0 (where $\cos kx = 1$ and $v_0' > 0$, a rarefaction) and towards its minima (where $\cos kx = -1$ and $v_0' = -ak^2 < 0$).

The caustic time. The supremum of $-v_0'$ is attained at the minima $x_m = (2m+1)\pi/k$ of T_0, so by Theorem VII.4(iv)

$$\mu_- = \sup_x\bigl(-v_0'(x)\bigr) = ak^2, \qquad t_c = \frac{1}{ak^2}. \tag{B3}$$

The characteristics $X(t;x_0) = x_0 + t\,v_0(x_0)$ from a neighbourhood of x_m converge on x_m, which is a fixed point of the flow, and meet there at t_c; for $t > t_c$ the map $x_0 \mapsto X(t;x_0)$ folds over near x_m and three characteristics pass through each point of a growing interval around it. The smaller the perturbation, the

Q^2	P^2	$d(Q,P)$ for $Q\cdot P=-2,-1,0,1,2,3,4$						
2	2	-209 304	130 329	50 064	25 353	648	327	0
4	2	-2 023 536	1 598 376	1 127 472	561 576	50 064	8 376	-648
6	2	-15 493 728	16 844 421	16 491 600	8 533 821	1 127 472	130 329	-15 600
4	4	-16 620 544	28 698 000	32 861 184	18 458 000	3 859 456	561 576	12 800
6	4	-53 249 700	474 507 528	632 078 672	392 427 528	110 910 300	18 458 000	1 127 472
6	6	2 857 656 828	11 890 608 225	16 193 130 552	11 232 685 725	4 173 501 828	920 577 636	110 910 300

TABLE IV. The index $d(Q,P)=(-1)^{Q\cdot P+1}c(Q,P)$ of quarter-BPS dyons of the heterotic string on T^6, from $1/\Phi_{10}$, in the chamber $\mathrm{Im}\,v<0$ (the expansion in powers of $e^{-2\pi i v}$). The table agrees entry by entry with Table 1 of [97]. Entries with $Q\cdot P<0$ are the other chamber's values at $|Q\cdot P|$ by the symmetry $v\mapsto -v$; their negativity for $Q\cdot P\le -1$ is the two-centred contribution of Theorem IV.2.

	$Q\cdot P=0$			$Q\cdot P=1$			$Q\cdot P=2$		
Q^2	$P^2=0$	$P^2=2$	$P^2=4$	$P^2=0$	$P^2=2$	$P^2=4$	$P^2=0$	$P^2=2$	$P^2=4$
0	2	64	816	0	0	-8	0	0	0
1	-12	-224	-1 248	1	96	1 968	0	0	-12
2	48	1 152	18 240	-8	-256	840	0	64	2 592
3	-168	-3 392	-10 320	37	1 376	34 656	-2	-224	2 432
4	528	11 520	200 736	-136	-3 840	16 632	16	1 152	43 392
5	-1 512	-30 336	-55 424	439	13 152	343 152	-72	-3 392	33 720
6	4 032	83 968	1 544 832	-1 288	-33 536	171 152	256	11 520	414 336
7	-10 176	-202 560	-179 022	3 503	92 928	2 476 752	-806	-30 336	302 400
8	24 528	496 512	9 480 000	-8 968	-220 672	1 265 256	2 320	83 968	2 926 080
9	-56 796	-1 118 496	-155 232	21 854	540 416	14 545 584	-6 200	-202 560	2 049 968
10	127 008	2 521 600	49 523 328	-51 080	-1 204 992	7 558 560	15 616	496 512	16 919 712
11	-275 544	-5 374 656	2 684 560	115 154	2 711 616	73 540 080	-37 508	-1 118 496	11 568 000
12	581 952	11 389 440	228 872 064	-251 528	-5 741 824	38 736 600	86 544	2 521 600	84 554 880

TABLE V. The index $d(Q,P)$ of quarter-BPS dyons in the $\mathbb{Z}_2$ toroidal orbifold of type IIB on T^6, from the product formula for $1/\widetilde{\Phi}_2$ with the twisted elliptic genera (9), in the chamber $\mathrm{Im}\,v<0$ and with the normalisation of Remark V.9. The entries agree with Tables 17, 18 and 19 of [13] in their common range $Q^2\le 8$. The column $P^2=0$ lies outside the single-centred cone; its sign is $(-1)^{Q^2}$. The negative entries inside the cone are those of Table III.

later the caustic, and no value of $a>0$ avoids it; the time is finite for every amplitude and every wavelength, and diverges only in the homogeneous limit $a\to 0$ or in the long-wavelength limit $k\to 0$ at fixed a.

The density. Along the characteristic from x_0,

$$\varepsilon\big(t,X(t;x_0)\big)=\frac{\varepsilon_0(x_0)}{1+t\,v_0'(x_0)},\qquad \varepsilon(t,x_m)=\frac{\varepsilon_0(x_m)}{1-t/t_c},$$

and the density at the minimum diverges as $(t_c-t)^{-1}$. Near the caustic, for $x_0=x_m+\xi$ with ξ small,

$$v_0'(x_0)=-ak^2\Big(1-\tfrac{k^2\xi^2}{2}(1+3a^2k^2)+O(\xi^4)\Big),$$

so at $t=t_c$ the linear term of $X(t_c;x_m+\xi)-x_m$ cancels and

$$X(t_c;x_m+\xi)=x_m+c\,\xi^3+O(\xi^5),\qquad c=\tfrac{k^2}{6}(1+3a^2k^2),$$

while the density there is $\varepsilon_0(x_m)\big/\big(\tfrac{k^2}{2}(1+3a^2k^2)\xi^2\big)\propto|x-x_m|^{-2/3}$: the integrable singularity of a fold caustic, with total energy in any interval finite at t_c and infinite density at one point. For $t>t_c$ the three-valued velocity field carries a finite energy density on each sheet and the dust description of [83] has no single-valued continuation.

Comparison with the full equation. Felder, Kofman and Starobinsky [41] integrated (11) numerically with the exponential potential and inhomogeneous initial data, and found that after a transient the solution is approximated by the free-streaming (Hamilton–Jacobi) solution, which is (12), and that the caustics predicted by it form in the full equation, with second derivatives of T growing without bound at the caustic; Barnaby [4] showed the same analytically for several effective actions. The dust limit therefore predicts the time of the first caustic, (B3) for the datum (B1), up to the duration of the transient during which $\partial T\cdot\partial T$ approaches -1; that transient is of the order of the time for the homogeneous pressure to become small, $x^0\sim$ a few string lengths by Theorem VII.1, and is independent of a and k. For a perturbation of wavelength $2\pi/k$ equal to a few string lengths and amplitude a of order one in string units, t_c is itself of order a string length, and the end state of the decay is never tachyon matter in any region of the brane.

DECLARATIONS

Data and code availability. The programs that compute the Fourier coefficients of Appendix A, the tables of this paper, the Lean 4 formalisation of its finite statements and the source

of the paper are in the repository github.com/creelie/paper_11, archived on Zenodo under doi:10.5281/zenodo.XXXXXXXX [7]. The catalogue of Section II is kept in the ledger format of the engine Anaxia [8]. Every reference of the bibliography was checked against its publisher, INSPIRE or Crossref record.

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